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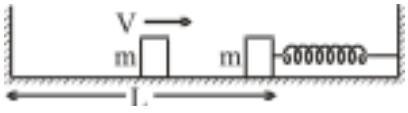
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PHYSICS

1) The left block in figure moves at a speed V towards the right block placed in equilibrium. All collisions to take place are elastic and the surfaces are frictionless. Find the time period of these periodic motions :

[Neglect the widths of the block] 

(1) $\pi\sqrt{\frac{m}{K}}+\frac{L}{V}$

(2) $\frac{\pi}{2}\sqrt{\frac{m}{K}}+\frac{L}{V}$

(3) $\pi\sqrt{\frac{m}{K}}+\frac{2L}{V}$

(4) $2\pi\sqrt{\frac{m}{K}}+\frac{2L}{V}$

2) **Assertion :** In SHM acceleration is always directed towards the mean position.
Reason : The body in SHM stops momentarily at the extreme positions and then moves back to mean position.

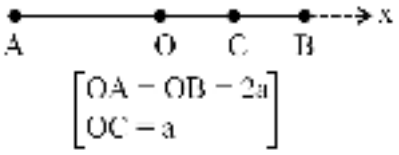
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

3) A particle of mass m is performing simple harmonic motion along line AB with amplitude 2a with centre of oscillation as O. At t = 0 particle is at point C (OC = a) and is moving towards B with



[OA = OB = 2a
OC = a]

velocity v = $a\sqrt{3}$ m/s. The equation of motion can be given by :

(1) x = 2a (sin t + $\sqrt{3}$cos t)

(2) x = 2a($\sqrt{3}$ sint + cos t)

(3) x = a (sin t + $\sqrt{3}$cos t)

(4) x = a ($\sqrt{3}$ sin t + cos t)

4) A simple pendulum has time period 2s. The point of suspension now moved upward according to relation y = (6t – 3.75 t^2) m where t is in second any y is the vertical displacement in upward direction. The new time period of simple pendulum will be :-

(1) 2s

(2) 1s

(3) 4s

(4) None of these

5) For a particle executing SHM along a straight line, match the statements in column-I with statement in column-II. (Note that displacement given in column-I is to be measured from mean position).

Column-I		**Column-II**	
(A)	Velocity-displacement graph may be	(p)	straight line
(B)	Acceleration-velocity graph may be	(q)	circle
(C)	Acceleration-displacement graph will be	(r)	ellipse
(D)	Acceleration-time graph will be	(s)	sinusoidal curve

(1) (A) q,r ; (B) q,r ; (C) p ; (D) s

(2) (A) s ; (B) q,r ; (C) p ; (D) q,r

(3) (A) p ; (B) q,r ; (C) q,r ; (D) s

(4) (A) p ; (B) s ; (C) q,r ; (D) q,r

6) Two pendulums of length 1.21 m and 1.0 m starts vibrating. At some instant, the two are in the mean position in same phase. After how many vibrations of the longer pendulum, the two will be in phase ?

(1) 10

(2) 11

(3) 20

(4) 21

7) A motion represented by equation $x = a\sin^3\omega t$ is :-

(1) Periodic and SHM

(2) Non periodic and SHM

(3) Periodic but no SHM

(4) Non periodic and no SHM

8) If in a stationary wave the amplitude corresponding to antinode is 4 cm, then the amplitude corresponding to a particle of medium located exactly midway between a node and an antinode is :-

(1) 2 cm

(2) $2\sqrt{2}$ cm

(3) $\sqrt{2}$ cm

(4) 1.5 cm

9) If velocity of sound in a gas is 360 m/s and the distance between a compression and the nearest rarefaction is 1m, then the frequency of sound is

(1) 90 Hz
(2) 180 Hz
(3) 360 Hz
(4) 720 Hz

10) The vibration of a string of length 60 cm fixed at both ends are represented by the equation $y = 2\sin\left(\frac{4\pi}{15}x\right)\cos(96\pi t)$ where x and y are in cm. The maximum number of loops that can be formed in it is :-

(1) 6
(2) 16
(3) 5
(4) 15

11) If we study the vibration of a pipe closed at one end and open at other end, then the following statement is not true :-

(1) Pressure change is maximum at closed end.
(2) Open end will be pressure antinode.
(3) Only odd harmonics of the fundamental frequency will be generated.
(4) Compression reflects as compression and rarefraction reflects as rarefraction at closed end.

12) A sonometer wire resonates with a given tuning fork forming standing waves with five antinodes between the two bridges, when a mass of 9 kg is suspended from the wire. When this mass is replaced by a mass M, the wire resonates with the same tuning fork forming three antinodes for the same positions of the bridges. The value of M is :-

(1) 25 kg
(2) 5 kg
(3) 12.5 kg
(4) (1/25) kg

13)

Decibel (dB) is the unit of :-

(1) intensity of light
(2) X-ray radiation capacity
(3) sound loudness
(4) energy of radiation

14) Regarding speed of sound in gas, match the following :-

	Column-I		Column-II
(A)	Temperature of gas is made 4 times and pressure 2 times	(P)	speed becomes $2\sqrt{2}$ times
(B)	Only pressure is made 4 times without change in temperature	(Q)	speed become 2 times
(C)	Only temperature is changed to 4 times	(R)	speed remains unchanged
(D)	Molecular mass of the gas is made 4 times	(S)	speed become half

(1) A – Q, B – R, C – Q, D – S

(2) A – Q, B – S, C – Q, D – S

(3) A – S, B –R, C – Q, D – S

(4) A – Q, B – R, C – Q, D – Q

15) A transverse wave is represented by the equation $y = y^0 \sin \frac{2\pi}{\lambda}(vt - x)$. For what value of λ, the maximum particle velocity equal to two times the wave velocity :

(1) $\lambda = 2\pi y_0$

(2) $\lambda = \frac{\pi y_0}{3}$

(3) $\lambda = \frac{\pi y_0}{2}$

(4) $\lambda = \pi y_0$

16) If the ratio of amplitude of two waves is 4 : 3, then the ratio of maximum and minimum intensity in their interference will be :-

(1) 16 : 18

(2) 18 : 16

(3) 49 : 1

(4) 1 : 49

17) An under water swimmer sends a sound signal to the surface. It produces 5 beats/sec when compared with fundamental tone of air column in a pipe of 20 cm length closed at one end. What is wavelength of sound in water.
(take, V_{water} = 1500 m/sec, V_{air} = 360m/sec)

(1) 3.3 m or 3.37 m

(2) 4.4 m or 4.47 m

(3) 2.5 m or 2.7 m

(4) 1m or 1.7 m

18) Two wires are fixed in a sonometer. Their tensions are in the ratio 8:1. The lengths are in the ratio 36:35. The diameters are in the ratio 4:1. Densities of the materials are in the ratio 1:2. If the higher frequency in the setting is 360 Hz, the beat frequency when the two wires sounded together is :

(1) 8
(2) 5
(3) 10
(4) 6

19) A vibrating tuning fork of frequency 500 Hz is brought at the upper end of resonance tube of length 1 m and velocity of sound is 330 m/s, then the total number of resonances obtained will be :

(1) 4
(2) 3
(3) 2
(4) 1

20) The amplitude of a damped oscillator becomes one-fourth in one minute. The amplitude after 3 minutes will be $\frac{1}{x}$ times of the original. Determine the value of x :-

(1) 12
(2) 256
(3) 4^3
(4) $\frac{1}{64}$

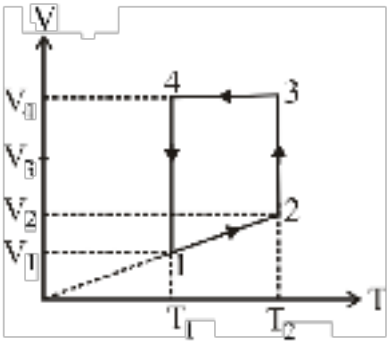
21) Equal amount of heat energy are transferred into equal mass of ethyl alcohol and water sample. The rise in temperature of water sample is 25°C. The temperature rise of ethyl alcohol will be (specific heat of ethyl alcohol is one half of the specific heat of water) :-

(1) 12.5°C
(2) 25°C
(3) 50°C
(4) It depends on the rate of energy transfer

22) The first two lengths of an air column in resonance tube experiment were found to be 32.1 cm and 99.2 cm respectively. Determine the end correction for the tube and frequency of the tunning fork (V_{sound} = 332 m/s)

(1) 1.45 cm, 260 Hz
(2) 1.45 cm, 270 Hz
(3) 2.45 cm, 247.4 Hz
(4) 1.45 cm, 247.4 Hz

23) A monoatomic ideal gas of two moles is taken through a cyclic process starting from point 1 as shown. $\frac{V_2}{V_1} = 2$ and $\frac{V_4}{V_1} = 4$. Temperature at point 1 is $T_1 = 27°C$. The temperature at point 2, T_2 is :



(1) 600 K

(2) 450 K

(3) 400 K

(4) 900 K

24) A block of mass 2.5 kg is heated to temperature of 500°C and placed on a large ice block. What is the maximum amount of ice that can melt (approx.) ?
[Specific heat for the body = 0.1 Cal/gm°C]

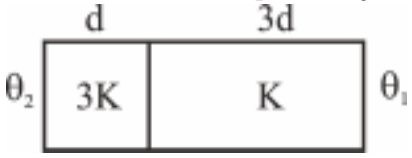
(1) 1 kg

(2) 1.5 kg

(3) 2 kg

(4) 2.5 kg

25) Two materials having coefficients of thermal conductivity '3K' and 'K' and thickness 'd' and '3d', respectively, are joined to form a slab as shown in the figure. The temperatures of the outer surfaces are 'θ_2' and 'θ_1' respectively, ($\theta_2 > \theta_1$). The temperature at the interface is :-



(1) $\frac{\theta_2 + \theta_1}{2}$

(2) $\frac{\theta_1}{10} + \frac{9\theta_2}{10}$

(3) $\frac{\theta_1}{3} + \frac{2\theta_2}{3}$

(4) $\frac{\theta_1}{6} + \frac{5\theta_2}{6}$

26) A vessel of volume $8.3 \times 10^{-3} m^3$ contains an ideal gas at temperature 27°C and pressure 200kPa. The gas is allowed to leak till the pressure falls to 100kPa and temperature increases to 327°C. What is the amount of gas in moles will be leaked out?

(1) $\frac{1}{2}$mol

(2) $\frac{1}{4}$mol

(3) $\frac{1}{8}$mol

(4) $\frac{3}{2}$mol

27) Two thermometers 'X' & 'Y' shows boiling point & freezing point of water as 220°X & 20°X and 120°Y & –40°Y respectively. If 'X' shows 100°X, then find the reading in 'Y' thermometer.

(1) 25°Y

(2) 50°Y

(3) 20°Y

(4) 24°Y

28) Two spheres of radii in the ratio 1 : 8 and densities in the ratio 4 : 1 and of same specific heat, are heated to same temperature and left in the same surrounding. Their rate of cooling will be in the ratio of-

(1) 1 : 1

(2) 1 : 2

(3) 2 : 1

(4) 1 : 4

29) For a gas, if $\frac{R}{C_P + C_V} = \frac{1}{6}$; then how much degrees of freedom will gas have ?

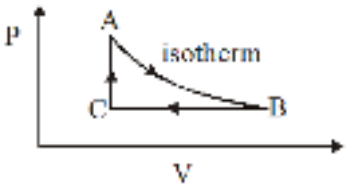
(1) 3

(2) 7

(3) 6

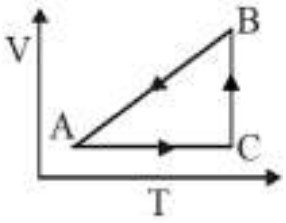
(4) 5

30) A cyclic process A–B→C→A is drawn on P-V diagram as shown.

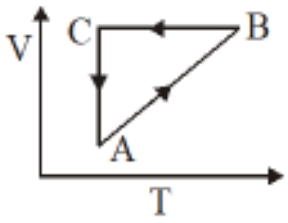


The cyclic process on V-T diagram would look like :

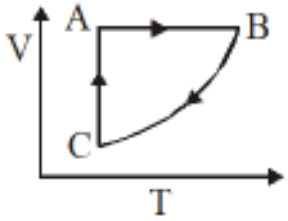
(1)



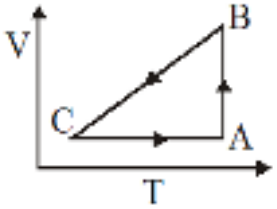
(2)



(3)



(4)



31) Steam at 100°C is passed into 22g of water at 20°C. The mass of water that will be present, when the water acquires a temperature of 90°C, (Latent heat of steam is 540 cal/g) is

(1) 24.8 g

(2) 24 g

(3) 36.6 g

(4) 30 g

32) Three discs A, B and C having radii 2m, 4m and 6m respectively are coated with carbon black on their outer surfaces. The wavelengths corresponding to maximum intensity are 300 nm, 400 nm and 500 nm, respectively. The power radiated by them are Q_a, Q_b and Q_c respectively, then :

(1) Q_a is maximum

(2) Q_b is maximum

(3) Q_c is maximum

(4) $Q_a = Q_b = Q_c$

33) Find the amount of work done by gas to increase the temperature of one mole of an ideal gas by 30° C, if it expands under the condition $V \propto T^{2/3}$:-

(1) 176 J

(2) 167 J

(3) 671 J

(4) 761 J

34) We took two separate gases with the same number densities for both. If the ratio of the diameters of their molecules is 4 : 1, then ratio of their mean free paths is :-

(1) 1 : 4

(2) 4 : 1

(3) 2 : 1

(4) 1 : 16

35) **Assertion (A)** :- The specific heat of a gas in an adiabatic process is zero and in a isothermal process in infinite.
Reason (R) :- Specific heat of gas is directly proportional to change of heat in system and inversely proportional to change in temperature.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

36) The ratio of temperatures of hydrogen and oxygen molecules at which their average speeds would be equal to the escape velocity from the surface of earth, is (Given that, escape velocity from earth's surface =11.2 km/s, Boltzmann constant, $k = 1.33 \times 10^{-23}$ J/K and Avogadro number, $N = 6.02 \times 10^{23}$ molecules/mole) :-

(1) 1/16

(2) 1/2

(3) 1

(4) 2/1

37) If two mole of monoatomic gas ($\gamma = 5/3$) is mixed with one mole of diatomic gas ($\gamma = 7/5$), the value of γ of mixture is :-

(1) 1.4

(2) 17/11

(3) 1.60

(4) 3.07

38) A simple pendulum, made of a bob of mass m and a metallic wire of negligible mass, has time period 2 s at T = 0°C. If the temperature of the wire is increased and the corresponding change in its time period is plotted against its temperature, the resulting graph is a line of slope S. If the coefficient of linear expansion of metal is α, then the value of S is :-

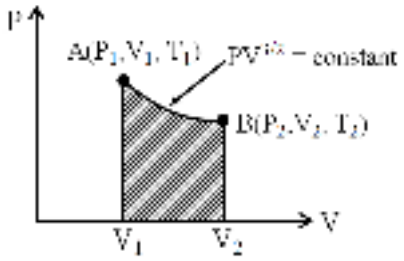
(1) $\frac{1}{\alpha}$

(2) 2α

(3) α

(4) $\frac{1}{2\alpha}$

39) The P-V diagram of path followed by one mole of perfect gas in a cylindrical container is shown in figure. The work done by gas, when the gas is taken from state A to state B, is :



(1) $2P_2V_1\left[1-\frac{\sqrt{V_2}}{\sqrt{V_1}}\right]$

(2) $2P_1V_1\left[1-\frac{\sqrt{V_1}}{\sqrt{V_2}}\right]$

(3) $2P_1V_2\left[\frac{\sqrt{V_1}}{\sqrt{V_2}}-1\right]$

(4) $2P_2V_2\left[1-\frac{\sqrt{V_1}}{\sqrt{V_2}}\right]$

40) A calorimeter contains 200 g of water at 10°C. Then 50 g of water at 100°C is further added and the temperature of water reaches to 27°C. The water equivalent of calorimeter is :

(1) 12.5 g

(2) 13.6 g

(3) 14.7 g

(4) 16.5 g

41) The amplitude of a particle executing SHM is made three-fourth keeping its time period constant. Its total energy will be :

(1) $\frac{E}{2}$

(2) $\frac{3}{4}E$

(3) $\frac{9}{16}E$

(4) None of these

42) **Statement-1 :** In the resonance tube experiment, if the tunning fork is replaced by another identical tuning fork but with its arms having been filed, the length of the air column should be increased to obtain resonance again.
Statement-2 : On filing the arms, the frequency of a tuning fork increases.

(1) Statement-1 is true, Statement-2 is true and Statement-2 is not the correct explanation of statement-1.

(2) Statement-1 is false, Statement-2 is true.

(3) Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation of Statement-1.

(4) Statement-1 is true, Statement-2 is false.

43) Velocity of sound waves in air is 330 m/sec. For a particular sound in air, a path difference of 40 cm is equivalent to a phase difference of 1.6π. The frequency of this wave is :-

(1) 165 Hz

(2) 150 Hz

(3) 660 Hz

(4) 330 Hz

44) Displacement-time equation of a particle executing SHM is $x = A \sin\left(\omega t + \frac{\pi}{6}\right)$. Time taken by the particle to go directly from $x = -\frac{A}{2}$ to $x = +\frac{A}{2}$ is :

(1) $\frac{\pi}{3\omega}$

(2) $\frac{\pi}{2\omega}$

(3) $\frac{2\pi}{\omega}$

(4) $\frac{\pi}{\omega}$

45) The energy of a simple harmonic oscillator in the state of rest is 3 joules. If its average kinetic energy is 8 joules, its total energy will be :-

(1) 16 J

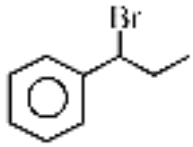
(2) 19 J

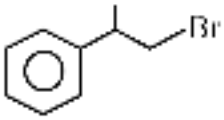
(3) 10 J

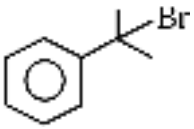
(4) 11 J

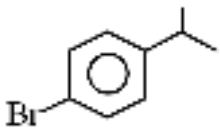
CHEMISTRY

1) Reaction of benzene with n-propyl chloride in the presence of aluminium chloride gives major product A. A on reaction with NBS gives product B. Then identify product B.

(1) 

(2) 

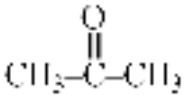
(3) 

(4) 

2) Controlled oxidation of ethane in presence of $(CH_3COO)_2Mn$ gives :

(1) $CH_3–CH_2–OH$

(2) $CH_3–CHO$

(3) 

(4) CH_3COOH

3) Which of the following is correct set of o,p-directing group with respect to electrophilic substitution reaction ?

(1) $–NO_2$, –CN, –CHO

(2) $–NHCOCH_3$, –Cl, –OH

(3) $–NO_2$, $–NH_2$, $–OCH_3$

(4) $–OCOCH_3$, –Cl, –CN

4) Propene can be converted to propane-1,2-diol by :

(1) $KMnO_4/H^{\oplus}/\Delta$

(2) O_3/H_2O

(3) H^+/H_2O

(4) dil. $KMnO_4$/273 K

5) Which of the following cannot be obtained in good yield by Wurtz reaction ?
(A) n-hexane
(B) 2,2-Dimethylpropane
(C) 2,3-Dimethylbutane
(D) 2-Methylbutane

(1) Only (B),(D)

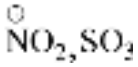
(2) Only (A), (C)

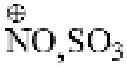
(3) Only (A), (D)

(4) Only (B), (C), (D)

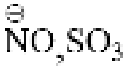
6)

Which of the following act as electrophile in the nitration and sulphonation of benzene?

(1) 

(2) 

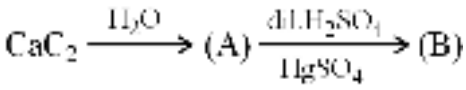
(3) HNO_3, H_2SO_4

(4) 

7) **Assertion :** The product of hydrogenation of 2-Butyne in the presence of Lindlar catalyst can

decolourise Br_2/CCl_4 solution.
Reason : Trans-2-butene is obtained by hydrogenation of 2-Butyne in presence of Lindlar catalyst.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

8) 

Identify product (B) in above reaction sequence.

(1) CH_3–CHO
(2) CH_3COCH_3
(3) CH_3–CH_2–OH
(4) CH_3–CH_3

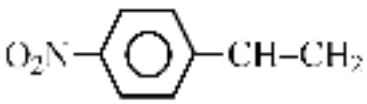
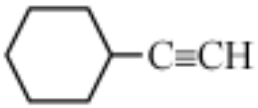
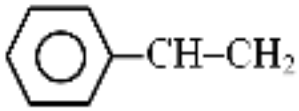
9) Which of the following reaction does not give benzene as product ?

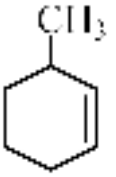
(1) Reaction of phenol with Zn dust
(2) Acidic hydrolysis of benzene sulphonic acid
(3) Heating propyne in red hot iron tube
(4) Reaction of n-hexane in presence of (Cr_2O_3 + Al_2O_3) at 773 K, 10-20 atm

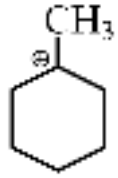
10) Chlorination of benzene in presence of UV light, 500 K temperature is an example of which reaction ?

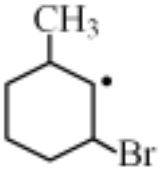
(1) Free radical substitution reaction
(2) Electrophilic substitution reaction
(3) Free radical addition reaction
(4) Electrophilic addition reaction

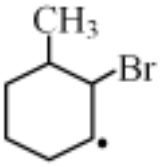
11) Which of the following most reactive for acidic hydration ?

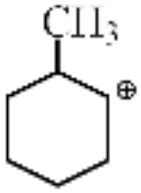
(1) 
(2) 
(3) CH_3–CH=CH_2
(4) 

12) The stable intermediate formed when  react with HBr in the presence of benzoyl peroxide :

(1) 

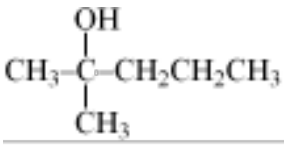
(2) 

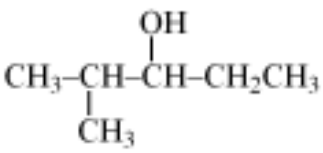
(3) 

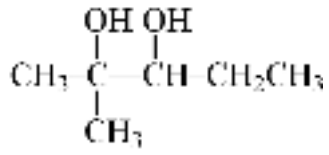
(4) 

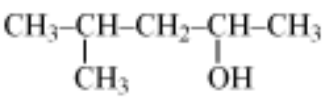
13)

An alkene (A) on ozonolysis gives a mixture of acetone and propanal. Then (A) on hydroboration oxidation gives which product ?

(1) 

(2) 

(3) 

(4) 

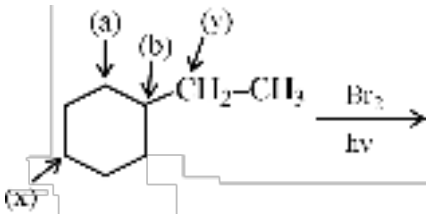
14) Which of the following is correct order of boiling point of alkane ?

(1) $CH_3(CH_2)_3CH_3 > (CH_3)_4C > CH_3(CH_2)_4CH_3 > CH_3CH_2CH_3$

(2) $CH_3CH_2CH_3 > CH_3(CH_2)_4CH_3 > (CH_3)_4C > CH_3(CH_2)_3CH_3$

(3) $CH_3(CH_2)_4CH_3 > CH_3(CH_2)_3CH_3 > (CH_3)_4C > CH_3CH_2CH_3$

(4) $CH_3(CH_2)_4CH_3 > (CH_3)_4C > CH_3(CH_2)_3CH_3 > CH_3CH_2CH_3$

15) 

In the above reaction hydrogen attached to which carbon is most likely to be substituted by bromine ?

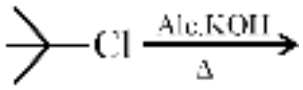
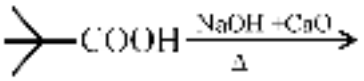
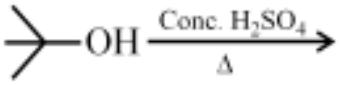
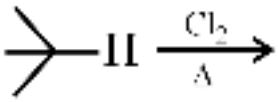
(1) a

(2) b

(3) x

(4) y

16) Match the column

	Column-I (Reactions)		**Column-II (Intermediate)**
A		p	Carbocation
B		q	Carbon free radical
C		r	Carbanion
D		s	No intermediate, only transition state

(1) A-r ; B-s ; C-p ; D-q

(2) A-p ; B-r ; C-s ; D-q

(3) A-s ; B-r ; C-q ; D-p

(4) A-s ; B-r ; C-p ; D-q

17) **Statement-I :** Ethylene chloride on heating with Zn gives ethene.
Statement-II : Reaction of vicinal dihalide with Zn is dehalogenation and with $NaNH_2$ is dehydrohalogenation.

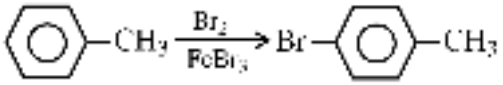
(1) Both Statement I and Statement II are correct.

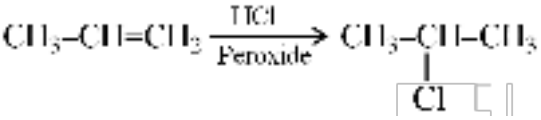
(2) Both Statement I and Statement II are incorrect.

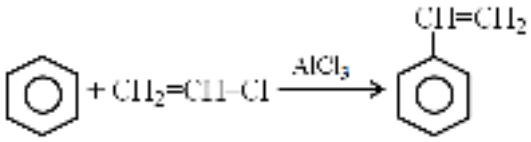
(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

18) Which of the following reaction not possible ?

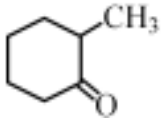
(1) 

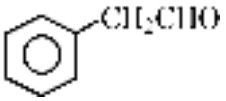
(2) 

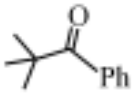
(3) 

(4) $CCl_3–CH=CH_2 \xrightarrow{HBr} CCl_3–CH_2–CH_2–Br$

19) Which of the following show tautomerism ?

(A) 

(B) 

(C) 

(D) 

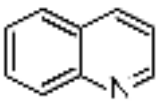
(1) Only A, B

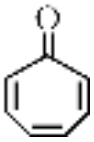
(2) Only A, B, D

(3) Only A, D

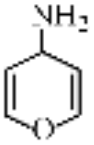
(4) Only B, C, D

20) Which of the following is not aromatic?

(1) 

(2) 

(3) 

(4) 

21) **Statement-I :** Electromeric effect can be observed in methyl cyanide and dimethyl ketone.
Statement-II : Inductive effect increases as the number of intervening bond increases.

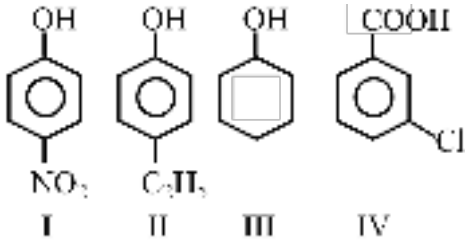
(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

22) What is the correct order of acidic strength of following compounds :



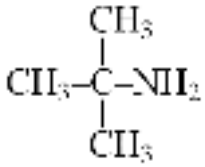
(1) I > IV > II > III

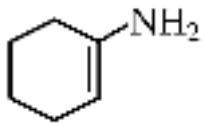
(2) IV > I > III > II

(3) I > II > IV > III

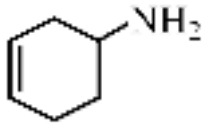
(4) IV > I > II > III

23) Among the following the compound with shortest carbon nitrogen bond length is :

(1) 

(2) 

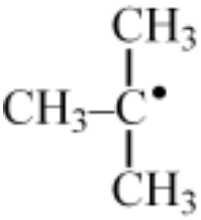
(3) $CH_2=CH-CH_2-NH_2$

(4) 

24) Which carbon free radical is least stable ?

(1) $(C_6H_5)_2\dot{C}H$

(2) $(C_6H_5)_3\dot{C}$

(3) 

(4) 

25) Which of the following is not an example of conjugated system ?

(1) Acrylaldehyde

(2) Vinyl chloride

(3) Allyl alcohol

(4) Pyrrole

26) Benzyl carbanion is less stable than :

(1) p-$CH_3C_6H_4CH_2^{\ominus}$

(2) p-$H_2NC_6H_4CH_2^{\ominus}$

(3) p-$O_2NC_6H_4CH_2^{\ominus}$

(4) p-$CH_3OC_6H_4CH_2^{\ominus}$

27) **Assertion :** Isobutylene is more stable than propene.
Reason : Molecular weight of isobutylene more than propene.

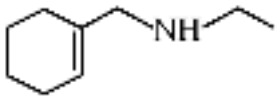
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

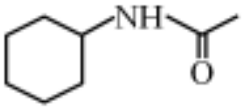
(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

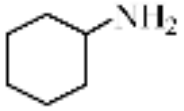
(3) Assertion is true but Reason is false.

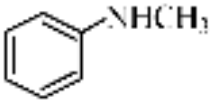
(4) Both Assertion and Reason are false.

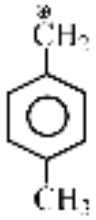
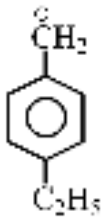
28) Which of the following will have maximum pK_b ?

(1) 

(2) 

(3) 

(4) 

29)  is more stable than  .
This can be explained by :

(1) Inductive effect

(2) Hyperconjugation

(3) Mesomeric effect

(4) Both (1) and (2)

30) Which statement is incorrect ?

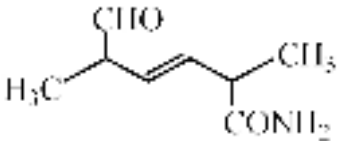
(1) –OH group show positive mesomeric effect but negative inductive effect.

(2) H-effect observed in $CH_3-\overset{\oplus}{C}H_2$ but not in $CH_3-\overset{\cdot}{C}H_2$

(3) Methyl carbocation and methyl free radical have incomplete octet of carbon.

(4) Ammonia and dichlorocarbene are neutral nucleophile.

31) The IUPAC name for the following compound is :



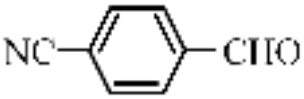
(1) 2,5-Dimethyl-6-carbamoylhex-3-enal

(2) 6-Formyl-2-methylhex-3-enamide

(3) 2,5-Dimethyl-5-carbamoylhex-3-enal

(4) 2,5-Dimethyl-6-oxohex-3-enamide

32) The number of π electrons and sp^2 carbon, respectively in the given compound is :



(1) 6, 7

(2) 12, 8

(3) 12, 7

(4) 10, 7

33) The molecular formula of higher homologue of acetic acid is :

(1) CH_2O_2

(2) C_3H_6O

(3) $C_3H_6O_2$

(4) $C_3H_8O_2$

34) Anisole, toluene and naphthalene are :

(1) Homocyclic benzenoid compound

(2) Homocyclic non-benzenoid compound

(3) Heterocyclic benzenoid compound

(4) Alicyclic compound

35) Which of the following known as valeraldehyde ?

(1) $CH_3CH_2CH_2COCH_3$

(2) $CH_3CH_2CH_2CH_2CH_2CHO$

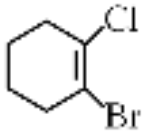
(3) $CH_3CH_2CH_2CH_2CHO$

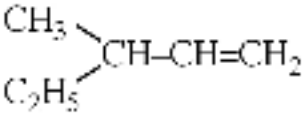
(4) CH_2=CH–CH_2CH_2CHO

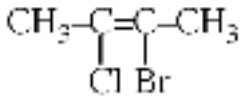
36) Which of the following IUPAC name is incorrect ?

(1) Cyclobutane carboxylic acid
(2) 2-Cyano cyclohexane carbaldehyde
(3) 2-Amino cyclohexanol
(4) 3-Cyanopropanoic acid

37) Which of the following can show geometrical isomerism?

(1) 

(2) 

(3) 

(4) 

38) Which of the following can show metamerism ?

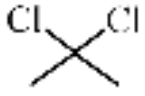
(1) n-butyl alcohol
(2) N-methyl ethanamine
(3) Benzyl methyl ether
(4) Methyl formate

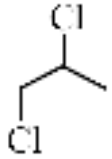
39) D-Erythrose and L-erythrose are related as :

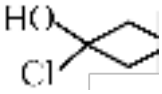
(1) Geometrical isomers
(2) Position isomers
(3) Diastereomers
(4) Enantiomers

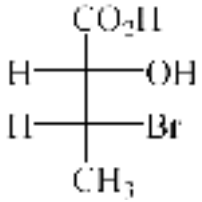
40) Which of the following show optical isomerism ?

(1) 

(2) 

(3) 

(4) 



41) The absolute configuration of given compound :

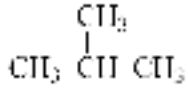
(1) (2R, 3R)

(2) (2R, 3S)

(3) (2S, 3R)

(4) (2S, 3S)

42) Match the column :

	Column-I (Compound)		**Column-II (Isomerism)**
A	CH_3CH_2OH	p	Geometrical isomerism
B		q	Functional isomerism
C	$CH_3–C\equiv C–CH_3$	r	Chain isomerism
		s	Position isomerism

(1) A-q ; B-r ; C-p

(2) A-q ; B-r ; C-q, s

(3) A-q ; B-r, s ; C-q

(4) A-p ; B-r ; C-q, s

43) The adsorbent used in adsorption chromatography is/are :
(A) Silica gel (B) Alumina
(C) Quick lime (D) Magnesia

(1) B only

(2) C and D only

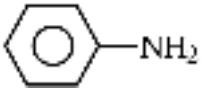
(3) A and B only

(4) A only

44) Which is not true for fractional distillation?

(1) Fractional distillation is used when difference in boiling point of two liquid is not much.

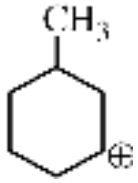
(2) In this technique vapours of liquid with higher boiling point condenses before vapours of liquid with lower boiling point.

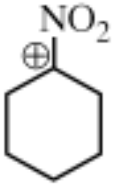
(3) $CHCl_3$ and  can be separated by fractional distillation.

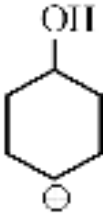
(4) Both (2) and (3)

45) Which of the following carbocation most stable?

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) In certain plants adapted to high temperature and light intensity, photosynthesis occurs via the C_4 pathway. In these plants, the initial fixation of carbon dioxide takes place in the mesophyll cells, where phosphoenolpyruvate (PEP) carboxylase fixes CO_2 into oxaloacetate, a 4-carbon compound. This compound is then converted into malate or aspartate and transported to bundle sheath cells, where decarboxylation releases CO_2 for the Calvin cycle. This spatial separation of primary and secondary CO_2 fixation helps in minimizing photorespiration and increases photosynthetic efficiency compared to C_3 plants.
Two statements are given with respect to above paragraph-
Statement-1 : In C_4 plants, the primary CO_2 fixation occurs in mesophyll cells by the action of PEP carboxylase.
Statement-2 : The main function of the C_4 pathway is to reduce photorespiration by maintaining a high concentration of CO_2 around RuBisCO in mesophyll cells.

(1) Both statements 1 and 2 are correct

(2) Both statements 1 and 2 are incorrect

(3) Statement 1 is correct, but statement 2 is incorrect

(4) Statement 1 is incorrect, but statement 2 is correct

2) From the statements given below choose the correct option :-
(A) In the photorespiratory pathway, there is neither synthesis of ATP nor NADPH.
(B) C_4 plants are known to be more tolerant to high temperature than C_3 plants.
(C) Photorespiration makes a major difference between C_3 and C_4 plant.
(D) More the ATP used in photorespiration more is the photosynthesis.
(E) RuBisCO has a lower affinity for CO_2 when the CO_2 : O_2 in nearly equal.

(1) A, B, D are true

(2) A, B, C are true

(3) B, C, D are true

(4) A, D, E are true

3) **Assertion :** Photorespiration decreases the efficiency of photosynthesis in C_3 plants.
Reason : During photorespiration, the enzyme RuBisCO acts as an oxygenase, leading to the consumption of O_2 and release of CO_2, which results in loss of carbon and energy.

(1) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true, and Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

4) (A) Photosynthesis is essentially a light dependent process, in which hydrogen from suitable oxidisable compound reduces the CO_2 into carbohydrates.
(B) Green substances in plants is located in special bodies in plant cells.
Name the researchers who stated the above statements.

(1) J.V. Sachs & Joseph priestly respectively

(2) T.W. Engelman & Van Neil respectively

(3) Van Neil & J.V. Sachs respectively

(4) Ingenhousz & J.V. Sachs respectively

5) Read the following statements
A. Within the chloroplast there is membranous system consisting of grana, the stroma lamellae, and the matrix, stroma .
B. There is not a clear division of labour within the chloroplast.
C. Usually the chloroplasts align themselves along the walls of the mesophyll cells, such that they get the optimum quantity of the incident light.
D. Photosynthesis does take place in the green leaves of plants
How many of the above statements are correct

(1) Four

(2) Three

(3) Two

(4) One

6) Which of the following statement is incorrect ?

(1) Antennae pigments help to make photosynthesis more efficient by absorbing different wavelengths of light.

(2) In PS II, reaction centre has absorption maxima at 680 nm.

(3) Electron transport system consist of cytochromes.

(4) Electrons excited from reaction centre of PS I are transferred to another acceptor molecule that has a lower redox potential.

7) **Assertion (A) :-** The reaction Centre is different in both the photosystems.
Reason (R) :- In PS-I the reaction Centre chlorophyll-a has an absorption peak at 700 nm, hence is called P700, while in PS-II it has absorption maxima at 680 nm and hence it is called P680.

(1) (A) is true but (R) is false

(2) (A) is false but (R) is true

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) Both (A) and (R) are true and (R) is not the correct explanation of (A)

8) **Assertion (A) :** Carboxylation is the most crucial step of the calvin cycle.
Reason (R) : Regeneration of the CO_2 acceptor molecule RuBP is crucial if the cycle is to continue uninterrupted.

(1) Both (A) & (R) are correct but R is not correct explanation of (A).

(2) Both (A) & (R) are correct and R is correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

9) Match list-I with list-II

	List-I		**List-II**
(A)	C_4 pathway	(I)	Sugar synthesis
(B)	C_3 pathway	(II)	ATP and NADPH synthesis
(C)	Light reaction	(III)	3-phosphoglycerate
(D)	Dark reaction	(IV)	Malic acid

Choose the correct answer from the options given below :

(1) A-IV, B-III, C-II, D-I

(2) A-IV, B-III, C-I, D-II

(3) A-I, B-II, C-III, D-IV

(4) A-III, B-II, C-I, D-III

10) Match the column-I with column-II and select correct option-

Column-I		**Column-II**	
(a)	Light reaction	(i)	Photorespiration

(b)	C_3 cycle	(ii)	Hatch and slack pathway
(c)	C_4 cycle	(iii)	z-scheme
(d)	C_2 cycle	(iv)	Calvin cycle

(1) a - ii, b - i, c - iii, d - iv

(2) a - i, b - iii, c - iv, d - ii

(3) a - iii, b - iv, c - ii, d - i

(4) a - iv, b - ii, c - i, d - iii

11) Given below are two statements :
Statement-I : Dark reaction does not directly depend on the products of the light reaction but dependent on the presence of light.
Statement-II : In biosynthetic phase ATP and NADH are used to drive the processes leading to synthesis of sugars.
In the light of above statement, choose the correct answer from the options given below :

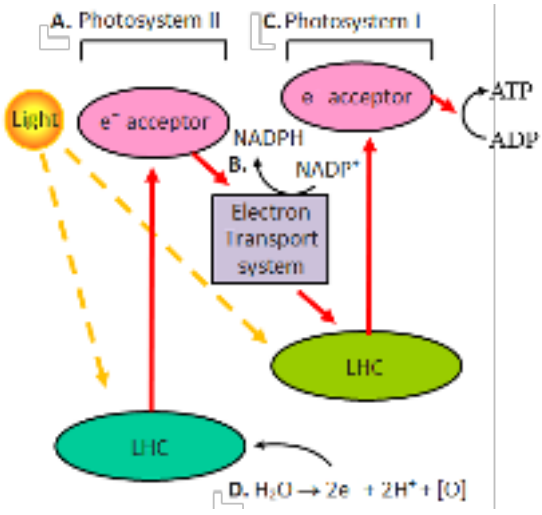
(1) Both **Statement I** and **Statement II** are incorrect.

(2) Both **Statement I** and **Statement II** are correct.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

12) Identify the point of error in the given figure :-



(1) Point A

(2) Point B

(3) Point C

(4) Point D

13) A biotechnology company is optimizing its industrial ethanol production process using yeast fermentation. During fermentation under anaerobic conditions, yeast converts glucose into ethanol, carbon dioxide, and other minor acidic byproducts. The process is monitored closely for pH changes, gas release, and substrate consumption. Despite controlled temperature and nutrient supply, occasional decreases in ethanol yield were observed.
Which of the following explanations best aligns with the observations in this fermentation system?

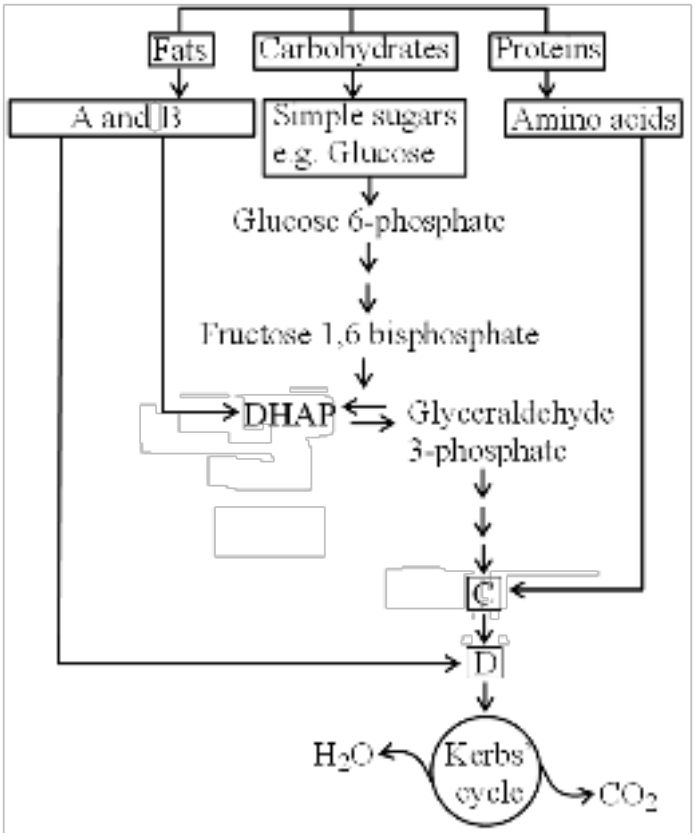
(1) Contamination by aerobic bacteria caused a shift to lactic acid fermentation, increasing alkalinity and reducing ethanol yield.

(2) Yeast switched to aerobic respiration due to oxygen contamination, producing water and carbon dioxide instead of ethanol.

(3) The glucose substrate underwent incomplete oxidation producing fatty acids and lowering carbon dioxide output.

(4) Yeast increased ethanol production under stress, causing heightened carbon dioxide and less acidic medium.

14) Below diagram shows the utilisation of different respiratory substrates. Identify A, B, C and D.



	A	B	C	D
(1)	Fru ctose-6-phosphate	Glyceral dehyde-3-phosphate	Fatty acid	Pyruvic acid
(2)	Fatty acid	Glycerol	Pyruvic acid	Acetyl-CoA
(3)	Glycerol	Fatty acid	Pyruvic acid	DHAP
(4)	Glycolic acid	Glycerol	Pyruvic acid	Acetyl-CoA

(1) 1

(2) 2

(3) 3

(4) 4

15) Read the following statements
A. Each plant part take care of its own gas-exchange needs.
B. Plants do not present great demands for gas exchange.
C. Distance that gases must diffuse even in large, bulky plants is not great.
D. There are sufficient reasons to believe that the first cells on this planet lived in an atmosphere that had oxygen.
How many of the above statements are correct

(1) Four
(2) Three
(3) Two
(4) One

16) Match the column-I and column-II and choose the correct option.

	Column-I		**Column-II**
A.	Alcoholic concentration during yeast poisoning	(i)	Less than 7%
B.	Energy in glucose released during fermentation	(ii)	Less than 1
C.	R.Q. of carbohydrate	(iii)	13% or more
D.	R.Q. of protein and fat	(iv)	1

(1) A-iii, B-i, C-iv, D-ii
(2) A-iii, B-iv, C-i, D-ii
(3) A-ii, B-iv, C-iii, D-i
(4) A-iv, B-iii, C-i, D-ii

17) Which of the following statement is incorrect :-

(1) Glycolysis occurs in the cytoplasm of cell
(2) In Kreb's cycle, there are three points where NAD^+ is reduced to NADH + H^+
(3) During Kreb's cycle Acetyl CoA is completely reduced into CO_2
(4) Plants unlike animal have no specialised organ for gaseous exchange.

18) **Statement-I :-** NADH is oxidised to NAD^+ rather vigrous in fermentation, however the reaction is very slow in aerobic respiration.
Statement-II :- In fermentation there is net gain of only four molecules of ATP for each molecule of glucose.

(1) Both **Statement I** and **Statement II** are false.
(2) **Statement I** is true but **Statement II** is false.
(3) **Statement I** false but **Statement II** is true.
(4) Both **Statement I** and **Statement II** are true.

19) Match the column A & column B and select the correct option-

Column A	**Column B**

(i)	Glycolysis	a	Formation of NADH + H^+, $FADH_2$ & ATP
(ii)	Link Reaction	b	Conversion of Pyruvic acid into acetyl Co-A
(iii)	Kreb's cycle	c	Formation of ATP from NADH + H^+ & $FADH_2$
(iv)	ETS	d	Partial breakdown of Glucose

(1) i-a, ii-b, iii-c, iv-d

(2) i-b, ii-c, iii-d, iv-a

(3) i-c, ii-d, iii-a, iv-b

(4) i-d, ii-b, iii-a, iv-c

20) **Assertion :-** The conversion of BPGA to 3-PGA in glycolysis is an energy releasing process.
Reason :- Two redox equivalents are removed from BPGA and transfered to NAD^+.
Select the correct answer from the options given below.

(1) Both Assertion and Reason are true and Reason is correct explanation by Assertion

(2) Assertion is true but Reason is false

(3) Both Assertion and Reason are false

(4) Both Assertion and Reason are true but Reason is not correct explanation of Assertion

21) Select incorrect statements-

(1) Cytochrome-C is an immobile carrier protein situated on inner membrane of mitochondria

(2) Cytochrome-C oxidase contain cytochrome a and a_3 along with two copper centres

(3) Complex-V is concerned with ATP synthesis

(4) Oxidation of one molecule of NADH give rise to 3ATP

22) What is the net gain of ATPs during complete oxidation of 2-molecule of Acetyl Co-A ?

(1) 12 ATP

(2) 30 ATP

(3) 24 ATP

(4) 6 ATP

23) How many total ATP would be formed through link reaction if one molecule of glucose is subjected to complete oxidation?

(1) 6 ATP

(2) 3 ATP

(3) 2 ATP

(4) 1 ATP

24) Inhibitory effect of oxygen on fermentation process is called as –

(1) Custer effect

(2) Crabtree effect

(3) Warberg's effect
(4) Pasteur effect

25) Respiratory pathway is an –

(1) Catabolic pathway
(2) Anabolic pathway
(3) Amphibolic pathway
(4) Endergonic pathway

26) How many net $NADH^+ + H^+$ are produced during alcoholic fermentation ?

(1) 2
(2) 4
(3) 1
(4) 0

27) Select the correct statement :-

(1) Glycolytic NADH transported to ETS by the shuttle system
(2) NADH produce in the Kreb's cycle transported to ETS by shuttle system
(3) In Kreb's cycle net ATP gain is 30
(4) In prokaryotes shuttle system is present

28) What goes in the calvin cycle to make, one molecule of glucose.

(1) $6CO_2$, 18 ATP, 12 NADPH
(2) 1 glucose, 18 ADP, 12 NADP
(3) 6 CO_2, 12 ATP, 18 NADPH
(4) 1 CO_2, 3ATP, 2 NADPH

29) To make one molecule of glucose how many turns of C_3 cycles are required :-

(1) Six
(2) Two
(3) One
(4) Twelve

30) Except for plants in shade or in dense forest, which of the following is rarely limiting factor of photosynthesis in nature ?

(1) CO_2
(2) H_2O
(3) Temperature
(4) Light

31) Other thylakoid pigment like chlorophyll b, Xanthophylls and catotenoids which are called __A__. Identify A ?

(1) A- chief pigment
(2) A-universal pigment
(3) A-accessory pigment
(4) A-normal pigment

32) In a chloroplast during day time the highest number of protons are found in :

(1) Stroma
(2) Lumen of thylakoid
(3) Inter membrane space
(4) Antennae complex

33) The PGR break seed and bud dormancy and synthesised in large amounts by tissue under going senescence and ripening of fruit is.

(1) Auxins
(2) Gibberellins
(3) Ethylene
(4) All

34) In which phase of growth cell attains maximal size in terms of wall thickening and protoplasm modifications ?

(1) Elongation phase
(2) Maturation phase
(3) Meristematic phase
(4) Both (1) & (3)

35) Which of the following meristem is responsible for primary growth in plants ?

(1) Vascular cambium
(2) Cork cambium
(3) Apical meristem
(4) Both (1) & (2)

36) Removal of shoot tips (decapitation) usually results in the growth of lateral buds. It is related to the removal of effect of which plant hormone :-

(1) Ethylene
(2) Auxin
(3) Cytokinin
(4) Gibberellic acid

37) _____ is the natural auxin.

(1) Indole acetic acid

(2) NAA (naphthalene acetic acid)

(3) 2,4-D (2,4-dichlorophenoxy acetic acid)

(4) Kinetin

38) Which of the following statements are incorrect.
A. ABA is a gaseous PGR.
B. Gibberellins induce internode elongation just prior to flowering in rosette plants (Beetroot and cabbage)
C. Removal of Roots is called decapitation
D. Ethylene is also known as stress hormone.
Choose the most appropriate incorrect answer from the options given below.

(1) A and B only

(2) A, B and C only

(3) A, C and D only

(4) A and D only

39) Match the column A with column B and select the correct option from given below :

	Column A		**Column B**
a	Auxins	i	Antagonist of GA_s
b	Gibberellines	ii	Overcome apical dominance
c	Cytokinins	iii	Speed up malting process in brewing industry
d	ABA	iv	Promote flowering in pineapple

(1) a-iv, b-iii, c-ii, d-i

(2) a-ii, b-iii, c-i, d-iv

(3) a-iv, b-ii, c-iii, d-i

(4) a-ii, b-iv, c-iii, d-i

40) Given below are statements one is labelled as assertion (A) and the other is labelled as reason (R).
Assertion (A) : Spray of Gibberellins can increase the sugarcane yield by as much as 20 tonnes per acre.
Reason (R) : Spraying sugarcane crop with gibberellins increases the length of the stem.
In the light of the above statements choose the correct answer from the options given below :

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

41) Given below are two statements :
Statement I : Auxin induces cell division, parthenocarpy, promotes flowering, cell enlargement and help in to initiate rooting in stem cutting.

Statement II : Auxin help to prevent fruit and leaf drop at early stages but promotes the abscission of older mature leaves and fruits.
In the light of the above statements choose the correct answer from the options given below:

(1) Both **Statement I** and **Statement II** are false.

(2) **Statement I** is true but **Statement II** is false.

(3) **Statement I** is false but **Statement II** is ture

(4) Both **Statement I** and **Statement II** are true.

42) Choose the correct set of option for size or weight of organ against time for the organism growing in natural environment.
(i) $W_1 = W_0 e^{rt}$
(ii) $L_t = L_0 + rt$
(iii) Linear growth curve
(iv) Sigmoid growth curve
(v) Arithmetic growth
(vi) Geometric growth

(1) (i), (iii), (iv)

(2) (ii), (iii), (iv)

(3) (i), (iv), (vi)

(4) (ii), (iv), (v)

43) **Statement-I :** Spraying juvenile conifers with cytokinins hastens the maturity period, thus leading to early seed production. Cytokinins also promotes bolting in beet, cabbage and many plants with rosette habit.
Statement-II : Auxin is one of the most widely used PGR in agriculture. The most widely used compound as source of auxin is ethephon.

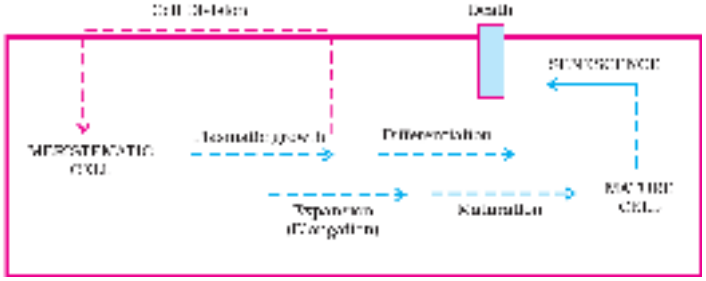
(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

44) Which of the following is not correct for given figure ?



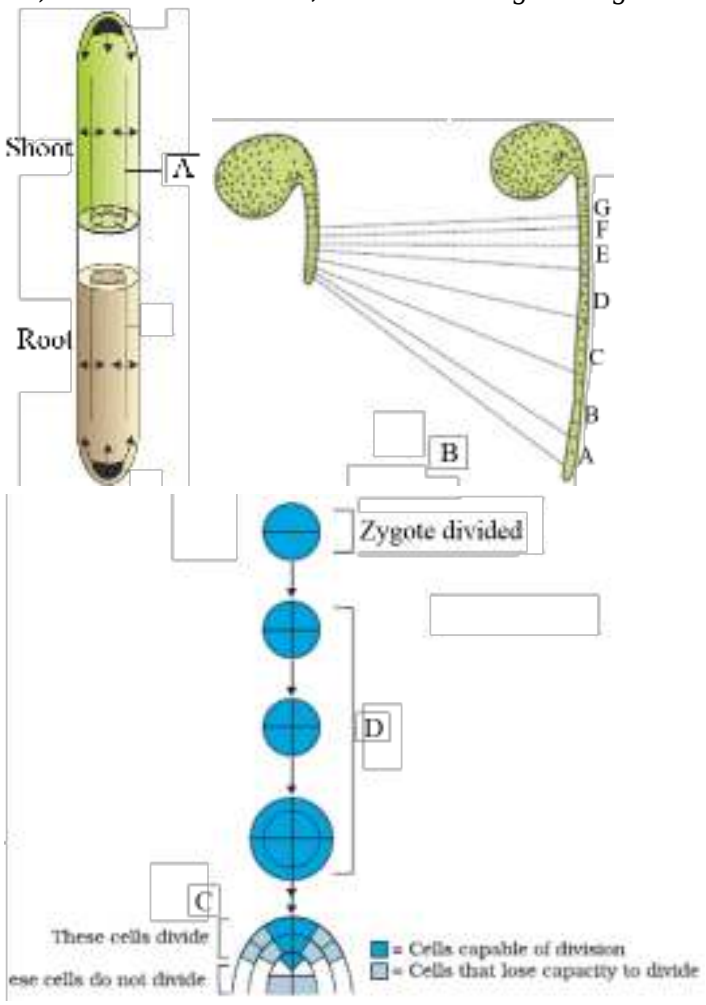
(1) It is representation of sequence of the developmental process in a plant cell.

(2) It is applicable to developmental processes in plant tissues.

(3) It is not applicable to developmental processes in plant organs.

(4) It shows the growth and differentiation in plant cell.

45) Find the labelled A, B & C & D in given figure.



	A	B	C	D
(1)	Vascular cambium	Parallel line technique	Geometric phase	Arithmetic phase
(2)	Cork cambium	Parallel line technique	Geometric phase	Arithmetic phase
(3)	Vascular cambium	Parallel line technique	Arithmetic phase	Geometric phase
(4)	Cork cambium	Relative growth	Arithmetic phase	Geometric phase

(1) 1

(2) 2

(3) 3

(4) 4

46) Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Visceral nervous system is the part of peripheral nervous system that comprises the whole complex of nerves, fibres, ganglia, and plexuses by which impulses travel from the central nervous system to the viscera and from the viscera to the central nervous system.

Reason (R) : The PNS is divided into two divisions called somatic neural system and autonomic

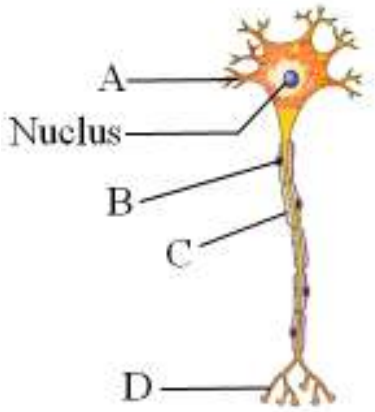
neural system.
In the light of above statements, choose the most appropriate answer from the options given below :

(1) Both (A) and (R) are correct and (R) is not the correct explanation of (A)

(2) (A) is not correct but (R) is correct

(3) (A) is correct but (R) is not correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)



47) Choose the correct function related to A, B, C and D :-

(1) A-help in internal conduction

(2) B-conduction of impulse towards cell body

(3) C-prevent leakage of ions

(4) D-Posses synaptic vesicles

48) Read following terminology of column-I and column-II and Identify their correct match :-

	Column-I		**Column-II**
(A)	Unipolar neuron	(i)	Sympathetic neural system
(B)	Bipolar neuron	(ii)	One axon and two or more dendrites
(C)	Multipolar neuron	(iii)	Found in retina of eye
(D)	Autonomic neural system	(iv)	Cell body with one axon
		(v)	One axon and one dendrites
		(vi)	Found in Embryonic stage
		(vii)	Found in cerebral cortex
		(viii)	Parasympathetic neural system

(1) A-(iv), (vi), B-(ii), (vii), C-(iii), (v), D-(i), (viii)

(2) A-(iv), (vi), B-(iii), (v), C-(ii), (vii), D-(i), (viii)

(3) A-(vii), (viii), B-(v), (vi), C-(ii), (iv), D-(i), (ii)

(4) A-(ii), (vii), B-(iii), (v), C-(iv)}, (vi), D-(i), (viii)

49) Given below are two statements :
Statement-I : Small amounts of androgenic steroids are also secreted by the adrenal cortex which play a role in the growth of axial hair, pubic hair and facial hair during puberty.
Statement-II : Aldosterone acts mainly at the renal tubules and stimulates the reabsorption of Na^+ and water and excretion of K^+ and phosphate ions and helps in the maintenance of electrolytes, body fluid volume, osmotic pressure and blood pressure.

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

50)

Match the Column I with Column II & Column III and choose the correct option

Column I		Column II		Column III	
(A)	Prolactin	(i)	Stimulate estrogen secretion	(a)	Lactation hormone
(B)	TSH	(ii)	Growth of mammary glands	(b)	Increased in exopthalmic
(C)	ACTH	(iii)	Stimulate cortisol secretion	(c)	Used in infertility treatment
(D)	FSH	(iv)	Stimulate thyroxin secretion	(d)	Regulated by CRH

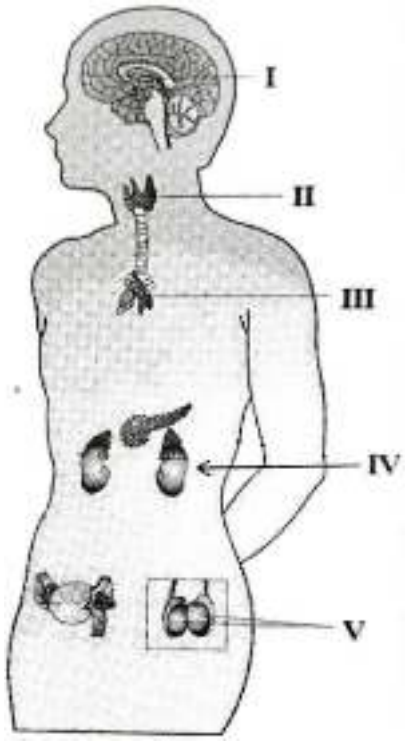
(1) A (ii) (c), B (iii) (d), C (i) (b), D (iv) (a)

(2) A (i) (d), B (ii) (a), C (iv) (b), D (iii) (c)

(3) A (ii) (a), B (iv) (b), C (iii) (d), D (i) (c)

(4) A (ii) (a), B (iv) (d), C (i) (b), D (iii) (c)

51) The given diagram represents the location of human endocrine glands I, II, III, IV and V.



I	Melatonin

II	Thymosin
III	Epinephrine
IV	Erthyropoietin
V	Testosterone

Which of the following gland is correctly matched with their secretions?

(1) I, II and III only

(2) I, IV and V only

(3) II, IV and V only

(4) II, III and V only

52) Read the following statements carefully and select the correct option :-
(A) Hormones exert their effects by binding to specific proteins called hormone receptors.
(B) Epinephrine is an example of a steroid hormone.
(C) Hormone receptor can be located on the cell membrane or inside the target cell.
(D) Hormones do not influence gene expression
(E) Membrane-bound receptors activate second messengers like cyclic AMP or IP3 to regulate cellular metabolism.
(F) Intracellular receptors, like those for steroid hormones, regulate gene expressions.

(1) All are correct

(2) Only B and D are incorrect rest are correct

(3) All are incorrect

(4) Only B and E are correct and A, C and D are incorrect

53) Examples of fibrous joint and cartilagenous joint is respectively :

(1) Joint between atlas and axis, joints between the carpals.

(2) Knee joint, skull bones.

(3) Skull bones, joint between adjacent vertebrae.

(4) Joint between the carpals, joint between atlas and axis.

54) Read the following statements (A-D) regarding smooth muscle.
(A) They are located in the inner walls of hollow visceral organs of the body like the alimentary canal, reproductive tract, etc.
(B) They do not exhibit any striation.
(C) They are called involuntary muscles.
(D) They assist for example, in the transportation of food through the digestive tract and gametes through the genital tract.
Select the correct option regarding statements A, B, C and D respectively where 'T' represent True and 'F' represent false.

(1) T, T, T, T

(2) F, T, F, T

(3) T, T, F, T

(4) F, T, T, T

55) Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Our vertebral column is formed by 26 serially arranged units called vertebrae and is dorsally placed.
Reason (R) : The number of cervical vertebrae are seven in almost all mammals including human beings.

(1) Both A and R true but R is not the correct explanation of A.

(2) A is true but R is false.

(3) A is false but R is true.

(4) Both A and R true but R is the correct explanation of A.

56) Match List-I with List-II.

	List-I		**List-II**
a.	Palm bones	i.	Femur
b.	Wrist bones	ii.	Metacarpals
c.	Ankle bones	iii.	Carpals
d.	Thigh bone	iv.	Tarsals

(1) a-iv, b-ii, c-iii, d-i

(2) a-i, b-iii, c-iv, d-ii

(3) a-i, b-iv, c-iii, d-ii

(4) a-ii, b-iii, c-iv, d-i

57) Given below are two statement :
Statement-I : Streaming of protoplasm in the unicellular organisms like amoeba is a complex form of movement.
Statement-II : Cytoskeletal elements like microfilaments are also involved in amoeboid movement.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-I is correct.

58) Which of the following statement is incorrect ?

(1) Each myosin filament is a polymerised protein.

(2) Many monomeric proteins called meromyosins constitute one thick filament.

(3) Each meromyosin has three important parts, a globular head, a cross arm and a tail.

(4) Globular head with a short arm called the heavy meromyosin.

59) A patient reports sleep disturbances and irregular body temperature patterns. On evaluation, the doctor suspects a hormonal imbalance affecting the body's 24 hour rhythm. The hormone involved is

secreted from a gland located on the dorsal side of the forebrain. Which hormone is most likely involved ?

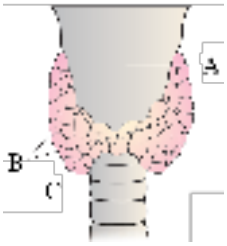
(1) Melatonin

(2) Cortisol

(3) MSH

(4) Thyroxine

60) In given diagram of identify A, B, C?



(1) A-Trachea, B-Vocal cord, C-Thyroid

(2) A-Vocal cord, B-Thyroid, C-Trachea

(3) A-Parathyroid, B-Trachea, C-Thyroid

(4) A-Trachea, B-Parathyroid, C-Thyroid

61) Read the following statements and choose the correct option
Statement A: Glucagon reduces the cellular glucose uptake and utilisation.
Statement B: Insulin stimulates conversion of glucose to glycogen (glycogenesis) in the target cells.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

62) **Statement-I:** Parathyroid hormone is responsible for demineralisation of bone.
Statement-II: Parathyroid hormone, decrease Resorption of bone calcium and increase reabsorption of calcium from renal tubules and digestive tract.
Read the following statements and select the right answer

(1) Statement-I is correct but statement-II is incorrect.

(2) Statement-I is incorrect but statement-II is correct.

(3) Both statement-I and statement-II is incorrect.

(4) Both statement-I and statement-II is correct.

63) Which of the following statements incorrect for thyroid gland ?

(1) The thyroid gland is composed two lobes which are located on either side of trachea.

(2) Due to cancer of thyroid gland or due to development of nodules of thyroid gland called hypothyroidism

(3) The thyroid gland is composed of follicles and stromal tissue.

(4) Thyroid follicular cells synthesized two hormone T_4 and T_3.

64) Which of the following statement is incorrect?

(1) Androgens act on the central neural system and influence the male sexual behaviour (libido).

(2) Progesterone is synthesised and secreted mainly by the growing ovarian follicles.

(3) Progesterone acts on the mammary glands and stimulates the formation of alveoli.

(4) JG cells of kidney produce a peptide hormone called erythropoietin which stimulates erythropoiesis.

65) **Assertion :** Deficiency of iodine in our diet result in hypothyroidism and enlargement of thyroid gland.
Reason : Iodine is essential for the normal rate of hormone synthesis in the thyroid.
Given below are assertion and reason mark the correct choice from given option :-

(1) Both **Assertion** and **Reason** are true but **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

66) Match List-I with List-II.

List-I		**List-II**	
(A)	Gastrin	(I)	Vasodilation
(B)	Erythropoietin	(II)	Pancreatic juice secretion
(C)	ANF	(III)	HCl secretion
(D)	Secretin	(IV)	RBCs production

Choose the correct answer from the option given below -

(1) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(2) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(3) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(4) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

67) Given below two statements choose the correct statement :-
Statement-I : Myelinated nerve fibres and unmyelinated nerve fibres both are enclosed by Schwann cells but Schwann cells form a myelin sheath only in myelinated nerve fibre.
Statement-II : Myelinated and unmyelinated nerve fibre are found in PNS.

(1) Both Statements are correct.

(2) Both Statements are incorrect.

(3) Statement-I correct and Statement-II incorrect.

(4) Statement-I incorrect and Statement-II correct.

68) Match these columns

Column-I	**Column-II**

(A)	CNS	(i)	Cranial and spinal Nerves
(B)	PNS	(ii)	Relay impulses from CNS to skeletal muscles
(C)	Somatic neutral system	(iii)	Transmits impulses from CNS to involuntary organs and smooth muscles
(D)	Autonomic neutral system	(iv)	Site of information processing and control

Codes :

	(A)	**(B)**	**(C)**	**(D)**
(1)	iv	iii	i	ii
(2)	iv	i	iii	ii
(3)	iv	i	ii	iii
(4)	iv	ii	iii	i

(1) 1

(2) 2

(3) 3

(4) 4

69) Match the brain regions with their associated functions :

	Column-I		**Column-II**
(A)	Pons	(i)	Fibre tracts connecting brain parts
(B)	Cerebellum	(ii)	Highly convoluted surface
(C)	Medulla oblongata	(iii)	Controls reflexes and gastric secretions
(D)	Brain stem	(iv)	Midbrain, pons, medulla

(1) A-(i), B-(ii), C-(iii), D-(iv)

(2) A-(ii), B-(i), C-(iv), D-(iii)

(3) A-(iii), B-(i), C-(ii), D-(iv)

(4) A-(i), B-(iii), C-(ii), D-(iv)

70) **Assertion :** Electrical synapse are faster than chemical synapse.
Reason : In electrical synapse, impulse transmission occurs through neurotransmitter.

(1) Assertion and Reason both false.

(2) Assertion true but reason false.

(3) Assertion and Reason both true.

(4) Assertion false but reason true.

71) Which of he following statements is correct ?

(1) The cerebral aqueduct runs through the pons.

(2) Corpora quadrigemina lies in the forebrain.

(3) Midbrain lies between thalamus and pons.

(4) Pons is a dorsal structure of midbrain.

72) Select the incorrect statement :-

(1) In resting stage the axonal membrane is comparatively more permeable to K^+ ions.
(2) In depolarisation stage sodium-potassium pump remain functional.
(3) In depolarisation inner axonal surface is positively charged.
(4) In polarisation stage outer axonal surface is positively charged.

73) Cerebral hemispheres are connected by a tract of nerve fibres called as -

(1) Corpus albicans
(2) Thalamus
(3) Corpus callosum
(4) Cerebellar peduncles

74) Which areas of the human brain are chiefly responsible for managing emotional reactions (such as fear, anger, pleasure) and regulation of sexual behavious ?

(1) Amygdala and medulla oblongata
(2) Limbic system and hypothalamus
(3) Cerebellum
(4) Pituitary gland

75) Which of the following contains a number of centres which control body temperature, urge for eating and drinking ?

(1) Thalamus
(2) Hypothalamus
(3) Amygdala
(4) Cerebellum

76) Which part of the brain controls respiration and cardiovascular reflexes ?

(1) Hypothalamus
(2) Cerebrum
(3) Medulla oblongata
(4) Cerebellum

77) Fill in the blank by selecting correct option ?
the dorsal portion of mid brain consists mainly of four round swelling called_____?

(1) Pons
(2) Medulla oblongata
(3) Corpora cavernosum
(4) Corpora quadrigemina

78) During movement of bones, force generated by ?

(1) Muscles
(2) Bones
(3) Joints
(4) Ligaments

79) Each coxal bone is formed by -

(1) Ilium, Ischium and Pubis
(2) Malleus, Incus and Stapes
(3) Thoracic, Lumbar and Cervical
(4) Tibia, Fibula, Patella

80) __Q__ is an auto immune disorder affecting neuromuscular junction leading to fatigue weakening and paralysis of skeletal muscle.

(1) Q = Tetany
(2) Q = Osteoporosis
(3) Q = Myasthenia gravis
(4) Q = Muscular dystrophy

81) Which structure in muscle fibres store calcium ions

(1) Sarcoplasma
(2) Sarcolemma
(3) Myofibril
(4) Sarcoplasmic Reticulum

82) Which of the following feature is not related with white muscles?

(1) Less myoglobin
(2) Less sarcoplasmic reticulum
(3) Less mitochondria
(4) depends on anaerobic respiration for ATP production

83) Male sexual behaviour is influenced by :-

(1) ACTH
(2) Oestrogen
(3) Testosterone
(4) Progesterone

84) Excess secretion of growth hormone in adults especially in middle age can result in severe disfigurement called?

(1) Gigantism
(2) Acromegaly
(3) Dwarfism
(4) Cretinism

85) The condition associated with diminished ability of the kidney to conserve water leading to water loss and dehydration is known as:

(1) Diabetes Insipidus
(2) Renal failure
(3) Cretinism
(4) Addison’s disease

86) The current scientific definition of hormone states that hormones are-

(1) Non-nutrient chemicals
(2) Intercellular messenger
(3) Produced in trace amounts
(4) All of the above

87) Emergency hormones perform all the given functions except-

(1) Increase alertness
(2) Pupillary constriction
(3) Piloerection
(4) Sweating

88) Fill in the blanks and choose the correct option
The secretion of PTH is regulated by the _____

(1) Circulating levels of TCT
(2) Thyroid stimulating hormone
(3) Circulating levels of calcium ions
(4) Autonomic nervous system

89) Select the odd one out w.r.t Hypothalamus

(1) Regulates a wide spectrum of body functions
(2) Contains several groups of neurosecretory cells
(3) Hormones produced by hypothalamus are of two types
(4) Forms & release prolactin

90) The correct sequence of events during contraction of muscle will be :-
(A) Action potential is generated in sarcolemma
(B) Ca^{++} binds with subunit of troponin
(C) Release of Ca^{++} ions into the sarcoplasm

(D) A neural signal reaching the neuromuscular junction releases a neurotransmitter-acetylcholine
(E) Removal of masking of active sites at actin for myosin
(F) Myosin head binds to exposed active sites on actin to form cross bridge.

(1) D, A, B, C, E, F

(2) D, C, A, B, E, F

(3) D, A, C, B, E, F

(4) A, B, C, E, F, D

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	2	4	3	1	1	3	2	2	2	2	1	3	1	4	3	1	3	2	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	4	1	2	2	1	4	3	4	4	1	2	2	4	2	1	2	3	2	3
Q.	41	42	43	44	45															
A.	3	2	3	1	2															

CHEMISTRY

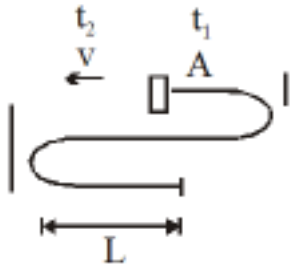
Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	4	2	4	1	1	3	1	3	3	4	3	2	3	2	4	1	3	2	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	4	2	4	3	3	2	2	2	4	4	3	3	1	3	2	3	3	4	3
Q.	86	87	88	89	90															
A.	3	2	3	3	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	2	1	3	2	4	3	1	1	3	1	2	2	2	2	1	3	1	4	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	3	1	4	3	4	1	1	1	4	3	2	3	2	3	2	1	3	1	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	1	3	3	1	3	2	4	3	2	2	3	1	1	4	4	3	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	1	2	2	1	2	1	3	1	2	3	2	3	2	2	3	4	1	1	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	2	3	2	1	4	2	3	4	3										

SOLUTIONS

PHYSICS



1)

Time period $T = t_1 + t_2$

$$t_1 = \pi\sqrt{\frac{m}{k}}$$

$$t_2 = \frac{L+L}{v} = \frac{2L}{v}$$

$$T = \pi\sqrt{\frac{m}{k}} + \frac{2L}{v}$$

Alternative

$T = t_1 + t_2$

t_1 = time taken to compress the spring and again release.

$$t_1 = \pi\sqrt{\frac{m}{k}}$$

t_2 = time taken to move towards wall and come again to spring.

$$t_2 = \frac{2L}{V}$$

$$\Rightarrow T = \pi\sqrt{\frac{m}{k}} + \frac{2L}{V}$$

2) Acceleration in SHM:

$\alpha = -\omega^2 x$

which shows it always points toward the mean position.

The assertion is correct because acceleration is always directed toward the mean position.

The reason is also true because the object stops momentarily at extremes before moving back.

However, the reason does not directly explain why acceleration is always directed toward the mean position.

3) $v^2 = \omega^2(A^2 - x^2) = 3\omega^2 a^2$

$$\Rightarrow \omega = \frac{v}{\sqrt{3}a} = \frac{\sqrt{3}a}{\sqrt{3}a} = 1 \text{ rad/sec}$$

$x = 2a \sin(\omega t + \phi)$, $v = 2a\omega \cos(\omega t + \phi)$

At $t = 0$, $x = a$ & $v = +ve$

$$\frac{1}{2} = \sin\phi \Rightarrow \phi = n\pi + (-1)^n \frac{\pi}{6} = \begin{cases} \pi/6 \\ 5\pi/6 \end{cases}$$

v at $\phi = \pi/6$ = + ve and v at $\phi = 5\pi/6$ = –ve

So, $\phi = \pi/6$

$x = 2a \sin(\omega t + \phi) = 2a[\sin \omega t \cos \phi + \cos \omega t \sin \phi]$

$x = a\left(\sqrt{3}\sin t + \cos t\right)$

4) $T = 2\pi\sqrt{\frac{\ell}{g}} = 2$

$y = 6t - 3.75t^2$

$\frac{dy}{dt} = 6 - 7.5t$

$a = \frac{d^2y}{dt^2} = -7.5\ m/s^2$

so point of suspension has acceleration in downward direction

so $g_{eff} = g - a$

$= 10 - 7.5 = 2.5$

$T = 2\pi\sqrt{\frac{\ell}{g-a}}$

$T = 2\pi\sqrt{\frac{\ell}{g-a}} = 2\pi\sqrt{\frac{\ell}{2.5}} = 4\,sec$

5)

(A) $v = \omega\sqrt{A^2 - x^2}$

$V^2 + \omega^2x^2 = \omega^2A^2$

(B) $V = \omega A \cos(\omega t)$

$a = -\omega^2 A \sin(\omega t)$

$\sin^2\omega t + \cos^2\omega t = 1$

$\Rightarrow \frac{a^2}{\omega^4A^2} + \frac{V^2}{\omega^2A^2} = 1$

(C) $a = -\omega^2 x$

(D) $a = -\omega^2 A \sin(\omega t)$

6) $nT_{long} = (n+1)\,T_{short}$

$\Rightarrow (n)\,2\pi\sqrt{\frac{1.21}{g}} = (n+1)\,2\pi\sqrt{\frac{1}{g}}$

$\Rightarrow \frac{11n}{10} = n + 1 \Rightarrow 11n = 10n + 10 \Rightarrow n = 10$

7)

$x = a\sin^3\omega t$

$x = a\left[\frac{3}{4}\sin\omega t - \frac{\sin 3\omega t}{4}\right]$

$x = \left[\frac{3a}{4}\sin\omega t - \frac{a}{4}\sin 3\omega t\right]$

So periodic but not SHM



8) $x=\lambda/8$

Let component waves are $y_1 = a \sin(kx - \omega t)$ & $y_2 = a \sin(kx + \omega t)$

 Resultant wave, $y = 2a \sin kx \cos \omega t$

$\Rightarrow$ Amplitude $= |2a \sin kx|$

 At antinode, amplitude $= 2a = 4$

 At required point, amplitude $= \left|2a \sin k \times \frac{\lambda}{8}\right|$

$= \left|2a \sin\left[\frac{2\pi}{8}\right]\right|$

$= \left|2a \sin \frac{\pi}{4}\right| = 4 \times \frac{1}{\sqrt{2}}$

$= 2\sqrt{2}\,\text{cm}$

9) Distance between a compression and the nearest rarefaction is $\frac{\lambda}{2} = 1\text{m}$.

Hence $n = \frac{\nu}{\lambda} = \frac{360}{2} = 180\,\text{Hz}$.

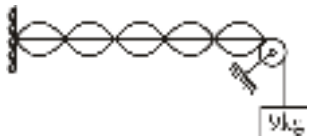
10) $k = \frac{2\pi}{\lambda} = \frac{4\pi}{15} \Rightarrow \lambda = \frac{15}{2}$

$\ell = p\left(\frac{\lambda}{2}\right) \Rightarrow p = \frac{2\ell}{\lambda} = \frac{2 \times 60}{15/2} = 16$

11) For COP :

closed end → displacement node (pressure antinode)

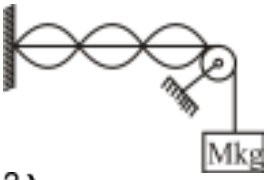
open end → displacement antinode (pressure node)



12)

$\frac{5\lambda}{2} = \ell$ $f_1 = \frac{v}{\lambda} = \frac{5v}{2\ell}$

$\Rightarrow f_1 = \frac{5}{2\ell}\sqrt{\frac{T}{\mu}}$ $\Rightarrow f_1 = \frac{5}{2\ell}\sqrt{\frac{9g}{\mu}}$



$\frac{3\lambda}{2} = \ell$ $f_2 = \frac{v}{\lambda} = \frac{3v}{2\ell}$

$f_2 = \frac{3}{2\ell}\sqrt{\frac{T}{\mu}}$ $\Rightarrow f_2 = \frac{3}{2\ell}\sqrt{\frac{Mg}{\mu}}$

$f_1 = f_2$ (resonance with T.F.)

$$\frac{5}{2\ell}\sqrt{\frac{9g}{\mu}} = \frac{3}{2\ell}\sqrt{\frac{Mg}{\mu}}$$

⇒ M = 25 kg

13)

Decibel is unit of sound loudness.

14)

Speed of sound, $V = \sqrt{\frac{\gamma RT}{M_w}}$

T → absolute temperature

M_w → Molecular mass

15) **Wave velocity:**

$$v = \frac{\omega}{k}$$

where, $\omega = \frac{2\pi v}{\lambda}$ and $k = \frac{2\pi}{\lambda}$, so

$$v = \frac{\frac{2\pi v}{\lambda}}{\frac{2\pi}{\lambda}} = v$$

Maximum particle velocity:

$v_{p,max} = \omega y_0$ From $v_{p,max} = \omega y_0$:

$\omega y_0 = 2v$

Substituting $\omega = \frac{2\pi v}{\lambda}$:

$$\frac{2\pi v}{\lambda} y_0 = 2v$$

Cancel v from both sides:

$$\frac{2\pi}{\lambda} y_0 = 2$$

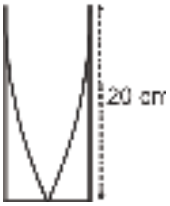
Solving for λ:

$$\lambda = \frac{2\pi y_0}{2} = \pi y^0$$

16) $\frac{I_{max}}{I_{min}} = \left(\frac{A_1 + A_2}{A_1 - A_2}\right)^2$

$= \left(\frac{4+3}{4-3}\right)^2 = 49 : 1$

17)



$n = \frac{v_{air}}{4\ell} = \frac{360}{4 \times 0.2} = 450$ Hz

Frequency of signal = 450 ± 5 = 445 Hz or 455 Hz

$\lambda_{water} = \frac{v_{water}}{n} = \frac{1500}{445} = 3.37\,m$ or $\frac{1500}{455} = 3.3\,m$

18) $f = \frac{1}{2L}\sqrt{\frac{T}{M/L}} = \frac{1}{2L}\sqrt{\frac{4TL}{\rho\pi D^2 L}}$

$= \frac{1}{2L}\sqrt{\frac{4T}{\pi D^2}}$

$\frac{f_1}{f_2} = \frac{L_2}{L_1}\sqrt{\frac{T_1}{T_2}\frac{\rho_2}{\rho_1}\frac{D_2}{D_1}} = \frac{35}{36}\sqrt{\frac{8 \times 2}{1 \times 1}} \times \frac{1}{4}$

$f_2 = 360$ Hz

So, $f_1 = 360 \times \frac{35}{36} = 350$ Hz

Beats = 360 – 350 = 10

19) $\lambda = \frac{v}{n} = \frac{330}{500} \times 100cm = 66cm$

1st resonance length $\ell_1 = \lambda/4 = \frac{33}{2} = 16.5$ cm

2nd resonance length $\ell_2 = 3\lambda/4 = 49.5$ cm

3rd resonance length $\ell_3 = 5\lambda/4 = 82.5$ cm

4th resonance not possible due to insufficient length of tube

20) Amplitude of damped oscillation is $A = A_0 e^{-rt}$

at t = 1 min, $A = \frac{A_0}{4}$, so, $\frac{A_0}{4} = A^0 e^{-r}$ or $e^r = 4$

after 3 min, $A = \frac{A_0}{x}$, So, $= e^{-3r}$

$\Rightarrow \frac{1}{x} = (e^r)^{-3} \Rightarrow x = (4)^3$.

21) $Q_{ethyl} = (Q)_{water}$

$m\frac{s}{2}\Delta T = m \times s \times 25$

$\frac{\Delta T}{2} = 25$

$\Delta T = 50°C$

22) $e = \frac{\ell_2 - 3\ell_1}{2} = \frac{99.2 - 3 \times 32.1}{2}$

= 1.45 cm

V = 2n(ℓ_2 – ℓ_1)

332 = 2n(99.2 – 32.1) $\times 10^{-2}$

n = 247.392 Hz

23) In 1 → 2, process is isobaric

$\frac{V_1}{T_1} = \frac{V_2}{T_2} \Rightarrow T_2 = \frac{V_2}{V_1}.T_1 = 2(27 + 273)$

= 600 K

24) msΔT = mL_f

$2.5 \times 10^3 \times (0.1)\ (500 - 0)$

$= M \times 80 \times 10^3$

$M = \frac{0.25 \times 500}{80}$

⇒ M ≈ 1.5 kg

25)

At steady state

$\frac{Q}{t} = \frac{3KA(\theta_2 - \theta)}{d} = \frac{KA(\theta - \theta_1)}{3d}$

On solving we get

$\theta = \frac{\theta_1}{10} + \frac{9\theta_2}{10}$

26)

PV = nRT

$(200 \times 10^3)\ (8.3 \times 10^{-3}) = nR\ (300)$ .....(1)

$(100 \times 10^3)\ (8.3 \times 10^{-3}) = n'R\ (600)$ .....(2)

Amount of mole leaked out

$n - n' = 8.3\left(\frac{2}{3R} - \frac{1}{6R}\right) = \frac{1}{2}$ mole

27) $\frac{T_X - (20°X)}{220°X - 20°X} = \frac{T_Y - (-40°Y)}{120°Y - (-40°Y)}$

$\frac{T_X - 20°X}{200°X} = \frac{T_Y + 40°Y}{160°Y}$

$\frac{100°X - 20°X}{200°X} = \frac{T_Y + 40°Y}{160°Y}$

$\frac{80°X}{200°X} \times 160Y = T_Y + 40°Y$

$T_Y = 24°Y$

28) $\left(\frac{dT}{dt}\right) = -\frac{\sigma eA\left(T^4 - T_s^4\right)}{ms}$

$\because R_F = \left(\frac{dT}{dt}\right) \propto \frac{A}{m} \propto \frac{r^2}{\rho r^3} \propto \frac{1}{\rho r}$

$\Rightarrow \frac{R_{F_1}}{R_{F_2}} = \frac{\rho_2 r_2}{\rho_1 r_1} = \frac{1}{4} \times \frac{8}{1} = \frac{2}{1}$

29) $\frac{R}{C_P + C_V} = \frac{1}{6}$

$\Rightarrow C_P + C_V = 6R$

& $C_P - C_V = R$

$C_V = 5R/2$

$\Rightarrow f = 5$

30)

A → B isothermal expansion, [T = Const.]

B → C isobaric compression [V ∝ T]

C → A isochoric process, P↑ → T↑ [V = Const.]

31) Heat loss = Heat gain

$m \times 540 + m \times 1 \times (100 - 90) = 22 \times 1 (90 - 20)$

m = 2.8 g

net mass of water in mixture = 22 + 2.8 = 24.8 g

32) $Q = e_r \sigma A T^4 t$

$P = \frac{Q}{t} = 1 \times \sigma A T^4$

$\Rightarrow P = \sigma A T^4$

$\Rightarrow P = \sigma(2\pi r^2)T^4$

$= \sigma(2\pi r^2)\left(\frac{b}{\lambda_m}\right)^4$

$\Rightarrow P \propto \left(\frac{r^2}{\lambda_m^4}\right)$

$Q_a : Q_b : Q_c = \frac{(2)^2}{(300)^4} : \frac{(4)^2}{(400)^4} : \frac{(6)^2}{(500)^4}$

$= \frac{4}{81} : \frac{1}{16} : \frac{36}{625}$

= (0.049) : (0.062) : (0.057)

33)

$VT^{-2/3} = const$

$V(PV)^{-2/3} = const$

$V^{1/3} P^{-2/3} = \text{const}$

$PV^{-1/2} = \text{const} \Rightarrow PV^x = \text{const}$

Polytropic process, $x = \dfrac{-1}{2}$

$$W = \frac{nR(\Delta T)}{1-x} = \frac{1 \times R \times 30}{1-\left(-\frac{1}{2}\right)} = \frac{2}{3} \times R \times 30 = 20 \times 8.31$$

$\simeq 167$ J

34) $\lambda_m = \dfrac{l}{\sqrt{2} n \pi d^2}$

$$\Rightarrow \frac{\lambda m_1}{\lambda m_2} = \frac{d_2^2}{d_1^2} = \left(\frac{1}{4}\right)^2 = \frac{1}{16} \quad (\because n_1 = n_2)$$

35) $S = \dfrac{Q}{m \cdot \Delta T}$ .....(1)

(I) For adiabatic process

$Q = 0$

from (1) $\boxed{S = 0}$

(II) For isothermal process

$T = \text{const.}$

$\Delta T = 0$

From (1) $\boxed{S = \infty}$

36) $v_{H_2} = v_{O_2}$, where, $v = \sqrt{\dfrac{8RT}{\pi M_w}}$

$$\sqrt{\frac{8RT_1}{\pi(2 \times 10^{-3})}} = \sqrt{\frac{8RT_2}{\pi(32 \times 10^{-3})}}$$

$$\Rightarrow \frac{T_1}{T_2} = \frac{1}{16}$$

37) $C_{v_1} = \dfrac{R}{\gamma - 1} = \dfrac{R}{5/3 - 1} = \dfrac{3R}{2}$

$$C_{v_2} = \frac{R}{7/5 - 1} = \frac{5R}{2}$$

$$\gamma_{mix} = \frac{1 \times \frac{7R}{2} + 2 \times \frac{5R}{2}}{2 \times \frac{3R}{2} + 1 \times \frac{5R}{2}} = \frac{17}{11}$$

38)

Variation of length of wire with temperature

$\Delta \ell = \alpha \ell \Delta \theta$ ... (i)

Now, time period of simple pendulum

$$T_\theta = 2\pi\sqrt{\frac{\ell + \Delta\ell}{g}}$$

$$= 2\pi\sqrt{\frac{\ell}{g}\left(1 + \frac{\Delta\ell}{\ell}\right)}^{1/2}$$

$$= T_0\left(1 + \frac{\Delta\ell}{2\ell}\right)$$

Variation in time period

$\Delta T = T_\theta - T_0$

$$= \frac{T_0\Delta\ell}{2\ell} = \frac{\alpha\ell\Delta\theta T_0}{2\ell}$$ [using (i)]

$$\Rightarrow \frac{\Delta T}{\Delta\theta} = \frac{\alpha T_0}{2} = \alpha \times \frac{2}{2} = \alpha$$

$\therefore S = \alpha$

39) For path A to B

$PV^{3/2}$ = constant = A (ray)

The work done

$$W = \int_{V_1}^{V_2} PdV = \int_{V_1}^{V_2} \frac{A}{V^{3/2}}dV$$

$$= A\left[\frac{V^{-1/2}}{-1/2}\right]_{V_1}^{V_2} = -2A\left[V^{-1/2}\right]_{V_1}^{V_2}$$

$$= -2A\left[\frac{1}{\sqrt{V_2}} - \frac{1}{\sqrt{V_1}}\right]$$

$$= 2A\left[\frac{1}{\sqrt{V_1}} - \frac{1}{\sqrt{V_2}}\right]$$

$$= 2P_1V_1^{3/2}\left[\frac{\sqrt{V_2} - \sqrt{V_1}}{\sqrt{V_1V_2}}\right]$$

$$= 2P_1V_1\left[1 - \frac{\sqrt{V_1}}{\sqrt{V_2}}\right]$$

40) Mass of cold water = 200 g

Mass of hot water = 50 g

Let w be the water equivalent of calorimeter.

Heat gained by water and calorimeter

$= (m_1 + w)(T - T_1)$

$= (200 + w)(27 - 10) = 17(200 + w)$ cal

Heat lost by hot water = (50)(1)(100–27) = 3650 cal

From principle of calorimetry,

Heat lost = Heat gained

$\Rightarrow (200 + w)(17) = 3650 \Rightarrow w = 14.7$ g

41) $E = \frac{1}{2}m\omega^2a^2$

$$\Rightarrow \quad \frac{E'}{E} = \frac{a'^2}{a^2} \quad \Rightarrow \quad \frac{E'}{E} = \frac{\left(\frac{3}{4}a\right)^2}{a^2} \left(\because a' = \frac{3}{4}a\right)$$

$$\Rightarrow \quad E' = \frac{9}{16}E$$

42) $f = \frac{V}{4\ell}$

$f \propto \frac{1}{\ell}$

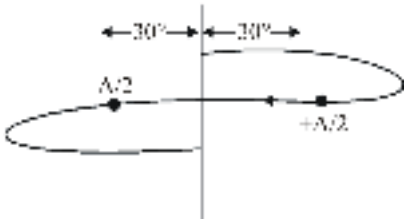
On filling, frequency of T.F. increases

43)

Phase difference = $\frac{2\pi}{\lambda}$ × path difference

$\Rightarrow 1.6\pi = \frac{2\pi}{\lambda} \times 40 \Rightarrow \lambda = 50 \text{ cm} = 0.5 \text{ m}$

$\Rightarrow v = n\lambda \Rightarrow 330 = 0.5 \times n \Rightarrow n = 660 \text{ Hz}$



44)

Time period of SHM = $\frac{2\pi}{\omega}$

 Time for traveling 60° is = $\frac{2\pi}{\omega}\frac{60}{360}$

$= \frac{\pi}{3\omega}$

45)

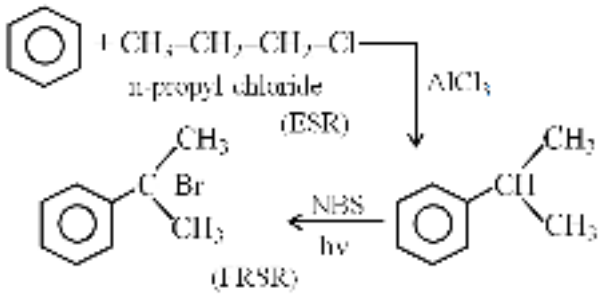
$<K.E.> = \frac{K.E_{max}}{2}$

$\Rightarrow K.E_{max} = 16 \text{ J}$

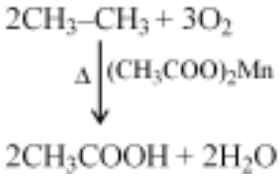
$\Rightarrow P.E_{min} = 3 \text{ J}$

Total energy = $KE_{max} + P.E_{min} = 19 \text{ J}$

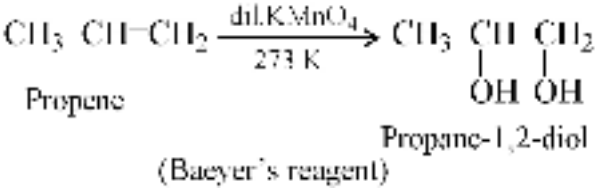
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46) 

(Note : $CH_3 - CH_2 - \overset{\oplus}{C}H_2$ rearrange to $CH_3 - \overset{\oplus}{C}H - CH_3$

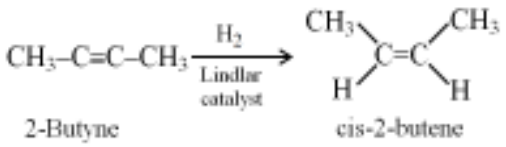
47) 

48) –$NHCOCH_3$, –Cl, –OH ⇒ All are +M group and o, p-directing.

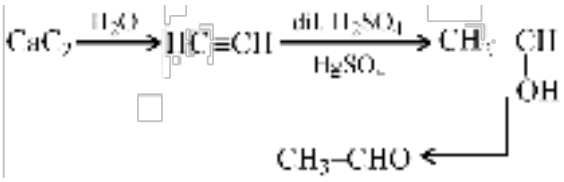
49) 

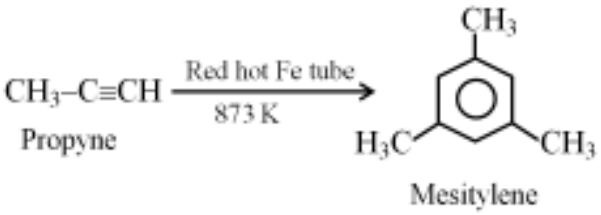
50) 2,2-Dimethylpropane and 2-Methylbutane are unsymmetrical and odd C alkanes
 Cannot be obtained in good yield by Wurtz reaction.

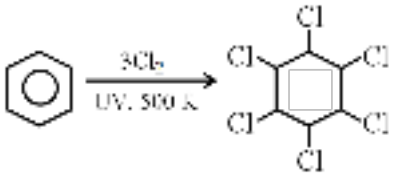
51) Electrophile in nitration of benzene ⇒ $\overset{\oplus}{N}O_2$
Electrophile in sulphonation of benzene ⇒ SO_3

52) 

Product is unsaturated cis-2-butene that gives test of unsaturation and decolourise bromine in CCl_4.

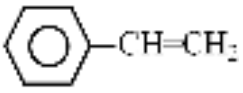
53) 

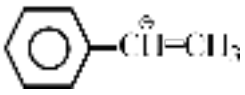
54) 

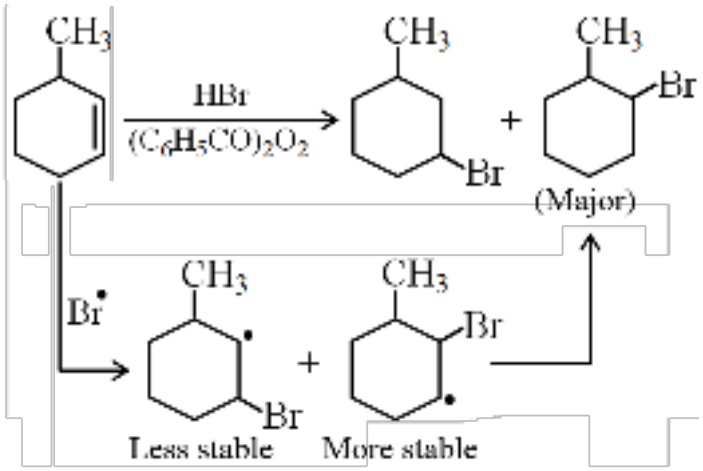
55) 

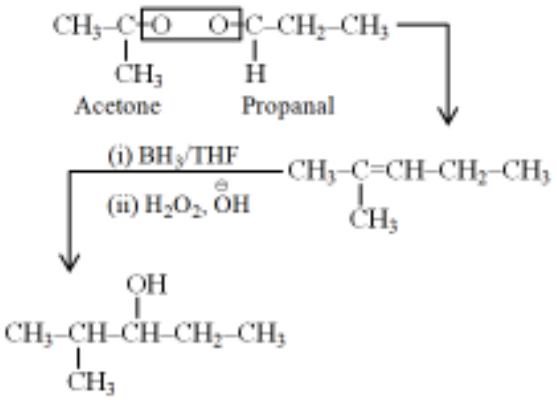
(Free radical addition reaction)

56) Reactivity towards hydration ∝ e^- density on C=C
∝ stability of carbocation



forms stable carbocation 

57) 

58) 

59) For alkanes,

Boiling point ∝ Molecular weight $\propto \frac{1}{\text{Branching}}$ (if mol. wt. same)

60) Above reaction is free radical substitution. Hydrogen attached to carbon (b) is most likely to be substituted because carbon free radical will be most stable.

61) Reaction (A) is E2 elimination in which no intermediate formed, only transition state.
Reaction (B) is sodalime decarboxylation in which carbanion intermediate formed.
Reaction (C) is E_1 elimination that involved carbocation intermediate.
Reaction (D) is FRSR that involve carbon free radical as intermediate.



62) ethene
Reaction of vicinal dihalide with Zn is dehalogenation, since halogen removed and with $NaNH_2$ is dehydrohalogenation. Since H and halogen both removed.

63)

Vinyl halide and aryl halide cannot be used as halide component in Friedal craft reaction with benzene.

64) (D) has hydrogen attached to sp^3 C at conjugate position.

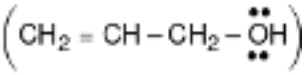
65) In (4) complete cyclic resonance is absent, so it is non-aromatic.

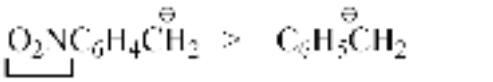
66) Methyl cyanide (CH_3–C≡N) and dimethyl ketone $(CH_3-\overset{\overset{O}{\|}}{C}-CH_3)$ has π-bond.
 can show electromeric effect.
Inductive effect decreases as the number of intervening bond increases.

67) General acidic strength order :
Carboxylic acid > Phenols > Alcohols
IV > I, II > III
I is more acidic than II due to –M of $–NO_2$
 Final acidic strength IV > I > II > III

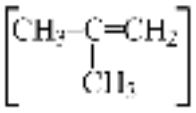
68) In (2), C–N bond has partial double bond character due to resonance so its bond length is minimum.

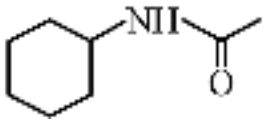
69) In (4), C radical is not stable by resonance, so it is least stable.

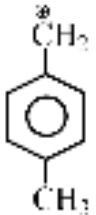
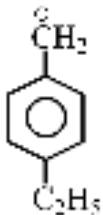
70) (1) Acrylaldehyde (CH_2=CH–CHO)
[π, π conjugation]
(2) Vinyl chloride (CH_2=CH–Cl)
(lone pair, π conjugation)
(3) Allyl alcohol 
(lone pair, π not in conjugation)
(4) Pyrrole 
(Lone pair, π conjugation)

71) 
(Benzyl carbanion)

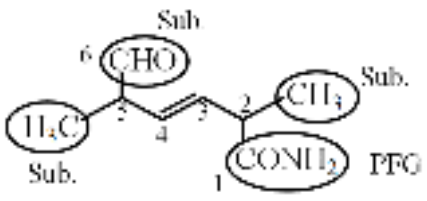
72)

Isobutylene  is more stable than propene (CH_3–CH=CH_2) because it has more number of α-hydrogen.

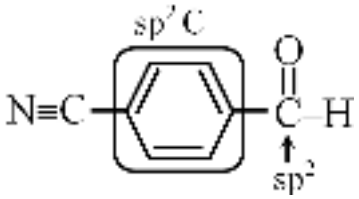
73)  is least basic so has maximum pK_b.

74)  is more stable than  because it has more number of α-H.
(H-effect of –CH_3 is greater than C_2H_5)

75) Dichlorocarbene (:CCl_2) is electrophile due to incomplete octet.

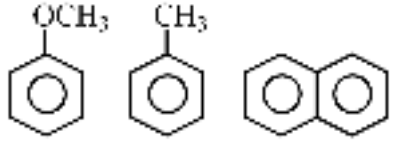


76) 2,5-Dimethyl-6-oxohex-3-enamide

77) 

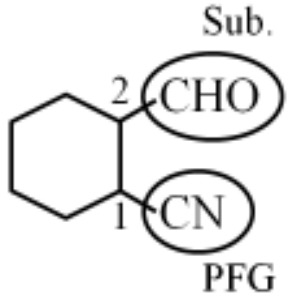
Total π bond = 6 ⇒ 12 π electrons
Total 7 sp^2 carbon

78) Acetic acid → CH_3COOH ($C_2H_4O_2$)
↓ $+CH_2$
Propanoic acid → CH_3CH_2COOH ($C_3H_6O_2$)
(Higher homologue)

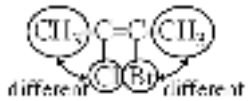
79) 

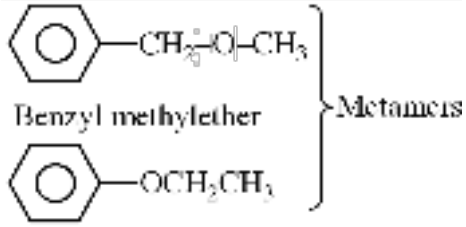
All are homocyclic benzenoid compounds.

80) Valeraldeyde ($CH_3CH_2CH_2CH_2CHO$)
(Valer means 5 carbon in saturated carbon chain including functional group)

81) 

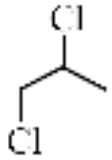
⇒ 2-Formyl cyclohexane carbonitrile (Correct)
2-Cyano cyclohexane carbaldehyde (Incorrect)

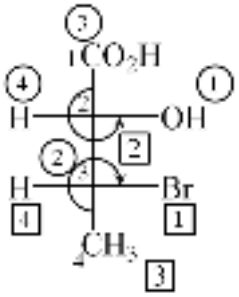
82)  GI
Geometrical isomerism

83) 

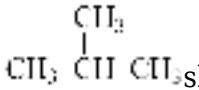
84)

D-Erythrose and L-erythrose are non-superimposable mirror image.
 Enantiomers

85)  contain 1 chiral carbon and show optical isomerism.

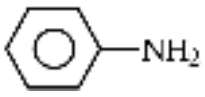
86) 
(2S, 3R)

87) CH_3CH_2OH show functional isomerism with
$CH_3–O–CH_3$

 show chain isomerism with
$CH_3–CH_2–CH_2–CH_3$
$CH_3–C\equiv C–CH_3$ show functional isomerism with $CH_2 = CH –CH=CH_2$
and position isomerism with $CH_3–CH_2–C\equiv CH$

88) Silica gel and alumina used as adsorbent in adsorption chromatography.

89)

$CHCl_3$ and  can not be separated by simple distillation due to large difference in there boiling point.

90)  is stable by resonance.

BIOLOGY

91) NCERT-XI, Pg. # 146

92) NCERT Pg. # 147

93) NCERT-XI, Pg. # 147

94) NCERT Pg.#134,135

95) NCERT-XI, Pg. # 136

96) NCERT, Pg. # 138, 139, 140

97) NCERT Pg. # 138

98) NCERT Page No.# 143

99)

NCERT Pg.#138,142,144,146

100) NCERT Pg. # 139, 144 to 146

101) NCERT Pg. # 142, 143

102) NCERT Pg. # 139

103) NCERT Pg # 157
It happened due to availability of oxygen and switching to aerobic respiration.

104) NCERT-XI, Pg. # 163

105) NCERT Pg # 154
Statement D is wrong.

106) NCERT-XI, Pg # 157, 163

107) NCERT Pg.# 158,159

108) NCERT Pg. # 161, 162

109) NCERT, Pg. # 168

110) NCERT Pg. # 156

111) NCERT Pg # 159-160

112) NCERT Pg. # 161

113) NCERT, Pg. # 158

114)

MODULE Pg. # 178

115) NCERT Pg. # 162

116) NCERT-XI, Pg. # 157

117) Module Pg. # 175

118) NCERT-XI, Pg # 145

119) NCERT-XI, Pg # 145

120) NCERT-XI, Pg. # 149

121) NCERT-XI, Pg # 138

122)

NCERT-XI, Pg. # 141

123) NCERT-XI, Pg # 177

124) NCERT-XI, Pg. # 169

125) NCERT-XI, Pg. # 169

126)

NCERT-XI, Pg. # 175, 177

127) NCERT Pg. # 175

128) NCERT-XI, Pg. # 175-178

129)

NCERT-XI, Pg. # 175-178

130) NCERT-XI, Pg. # 176

131) NCERT-XI, Pg. # 175

132) NCERT Pg. # 170

133) NCERT Pg. # 176, 177

134) NCERT-XIth , Pg. # 173

135) NCERT Pg. # 168, 169

136) NCERT (XII) ; Page No. 231

137) NCERT Pg. # 317

138) NCERT Pg. # 232

139) NCERT-XI, Pg. # 245

140) NCERT, Pg # 241, 242

141)

NCERT-XI, Pg. # 240

142) NCERT-XII, Pg. # 247-248

143) NCERT Pg # 227

144)

NCERT Pg.# 218

145) NCERT Pg.#225

146) NCERT Pg.#25,26

147) NCERT XI Pg.# 217, 218

148) NCERT Pg.#221.

149) NCERT XI Pg.#242

150) NCERT Pg.#243

151) NCERT, Pg 245, 246

152) NCERT, Pg. # 243

153) NCERT Pg.#243

154) NCERT, Pg # 246, 247

155) NCERT Pg.#243

156) NCERT Pg. # 247

157) NCERT Pg.#232

158) NCERT (XII) ; Page No. # 316 (E)
NCERT (XII) ; Page No. # 316 (H)

159) NCERT XI Pg.#236

160) NCERT Pg. No. # 234

161) NCERT XI Pg.#236

162) NCERT, Pg. # 232-33

163) NCERT (XII) ; Page No. # 321 (E)
NCERT (XII) ; Page No. # 320 (H)

164) NCERT Pg. # 236

165) NCERT Pg. # 236

166) NCERT Pg. No. # 236

167) NCERT Pg. 236

168) NCERT Pg.#227

169) NCERT Pg.#24,25,26

170) NCERT Pg.#227

171) NCERT XI Pg.#219

172) NCERT Pg. # 308

173) NCERT 337

174) NCERT Pg. No. : 241

175) NCERT Page No. #242

176) NCERT Pg # 239

177) NCERT Pg # 244

178) NCERT, Pg # 243

179) NCERT, Pg # 240

180)